وزارة الصحة
Ministry of Health
المديرية العامة للخدمات الصحية بمحافظة شمال الشرقية
دائرة التخطيط والتنظيم الصحي
قسم إدارة المعلومات الصحية

Waiting time report on referrals to Ibra Hospital, year 2025

April 2026

Report Summary

This report analyzes referral waiting time and response time for patients referred to Ibra Hospital in 2025, using data from the Nabd Al-Shifa system. It covers major departments including Cardiology, Dermatology, ENT, General Medicine, Surgery, Obstetrics & Gynecology, Ophthalmology, Orthopedics, and Pediatrics, and includes both routine and urgent referrals.

Overall, the findings show that:

- Routine referrals consistently have longer waiting times than urgent referrals, indicating that prioritization mechanisms are generally functioning.
- There is significant variation between departments:
  - Ophthalmology and Orthopedics have the longest waiting times and highest variability.
  - ENT and Cardiology show moderate delays with many extreme cases.
  - General Surgery, Pediatrics, and Obstetrics & Gynecology perform better with shorter waiting times.
- Outliers (extremely long waits) are present in nearly all departments, suggesting inconsistencies in scheduling and possible backlog issues.
- Response times (decision-making time) are generally short across departments, with most cases processed within a few days, although some extreme delays still occur.
- Appointment slot utilization ranges between 72% and 83% in most departments, indicating relatively good use of available capacity, with some inefficiencies in certain areas like General Surgery.

Introduction

Referral waiting time is a key indicator of healthcare system efficiency and service quality. Timely referrals enable patients to access appropriate specialist care promptly, which can improve health outcomes and reduce the risk of complications. Monitoring referral waiting times also helps healthcare institutions identify service gaps and improve care coordination.

Objective

This report presents the analysis and discussion of referral waiting time data for Ibra Hospital in 2025.

Data

The selected data represents all approved referrals to Ibra Hospital that were given appointments in the year 2025. The source of this data is the Nabd Al-Shifa program. It includes data from departments with higher referral frequencies, namely: Cardiology (Adult), Dermatology, ENT, General Medicine, General Surgery, Obstetrics & Gynecology, Ophthalmology, Orthopaedics, and Paediatrics. The types of referrals included were both routine and urgent.

Terms

- Response Time: The duration between the date a patient is referred and the date a decision is taken, either by accepting the case (approved) or otherwise (cancelled, rejected, held, etc.).
- Waiting Time: The duration between the date a patient is referred and the date of the appointment given.

Data Analysis

Descriptive statistics were used in the data analysis through the following techniques:

- Box Plot

A box plot summarizes the distribution of the data. The line inside the box represents the median, which is the middle value of the dataset. The box shows the middle 50% of the observations, indicating where most of the data is concentrated. The lines extending from the box, called whiskers, indicate the minimum and maximum values that are not considered extreme. Any individual points outside the whiskers represent outliers, which are unusually high or low values.

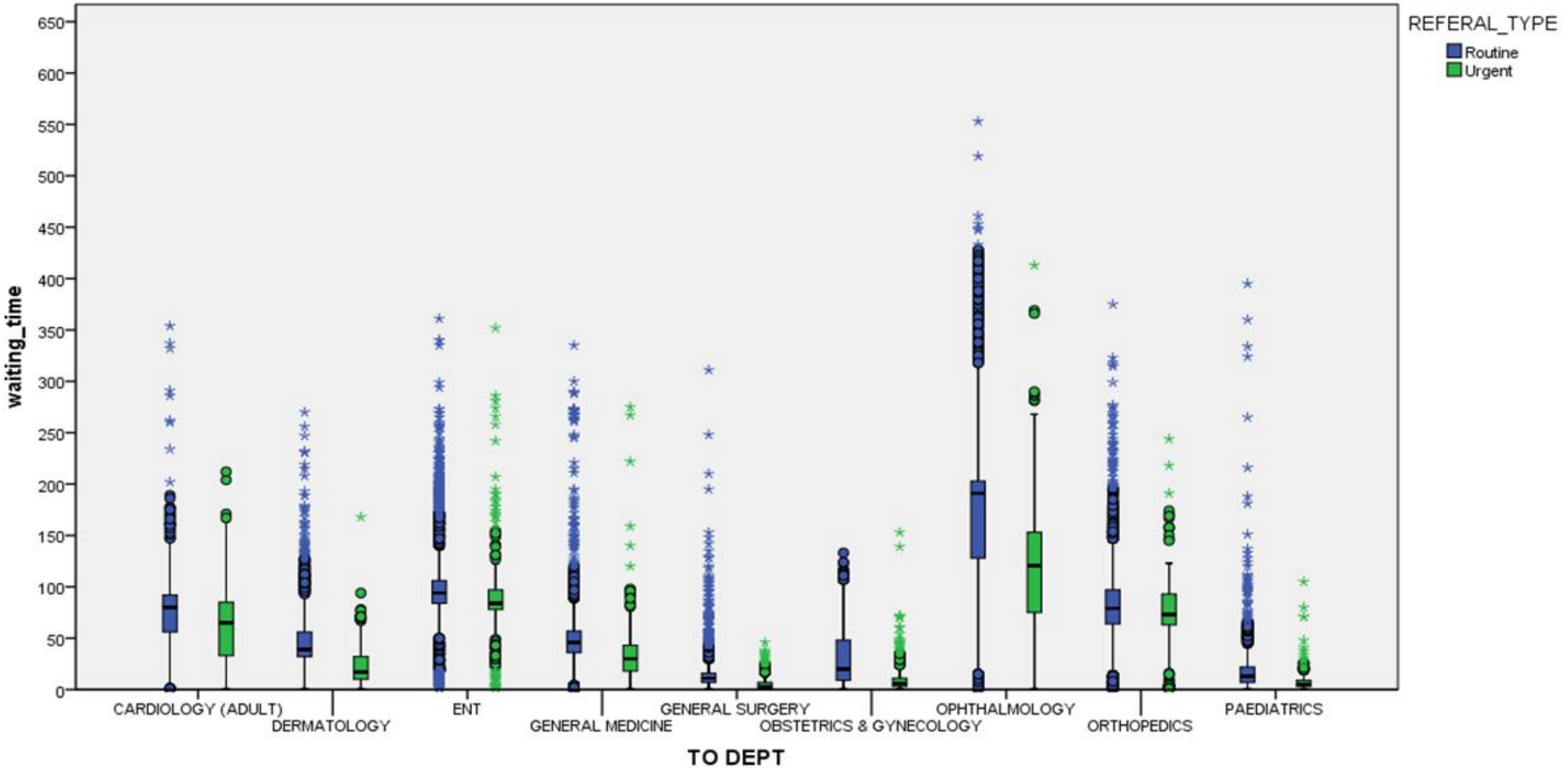
- Bar Graph
- Cumulative Percent Line Graph

A cumulative percent line graph shows the proportion of patients who were seen within a given waiting time. Each point represents the percentage of patients who waited up to that specific number of weeks. The curve gradually increases until it reaches 100%, representing all patients in the dataset.

- Median

The median is the middle value of a dataset that divides the data into two equal parts. It was used instead of the mean to show the typical or central value of a dataset since it is not affected by extreme values (outliers).

Section 1
waiting time overall view


REFERAL_TYPE
Routine
Urgent
waiting_time
650
600
550
500
450
400
350
300
250
200
150
100
50
0
CARDIOLOGY (ADULT)
DERMATOLOGY
ENT
GENERAL MEDICINE
GENERAL SURGERY
OBSTETRICS & GYNECOLOGY
OPHTHALMOLOGY
ORTHOPEDICS
PAEDIATRICS
TO DEPT

The figure is a boxplot comparing referral waiting time across hospital departments, separated by referral type (Routine vs Urgent).

Overall pattern

Routine referrals (blue) generally have longer waiting times than Urgent referrals (green) in most departments.

This pattern suggests that the referral system is prioritizing urgent cases, which is expected in a functioning triage system.

Departments with the longest waiting times

Ophthalmology shows the highest waiting times, especially for routine referrals.

> The median appears around 150–200 days, with some cases exceeding 500 days.

Orthopedics also shows relatively long waiting times, particularly for routine referrals.

These departments may have high demand, limited specialists, or capacity constraints.

Departments with moderate waiting times

Departments such as:

- ENT
- Cardiology (Adult)
- General Medicine

show moderate waiting times, with routine referrals generally around 40–100 days.

Departments with shorter waiting times

- General Surgery
- Obstetrics & Gynecology
- Pediatrics

These departments show shorter median waiting times, particularly for urgent referrals, suggesting better service flow or lower demand pressure.

Variability (spread of waiting time)

Some departments show large variability and many outliers (points above the whiskers), especially:

- Ophthalmology
- ENT
- Orthopedics

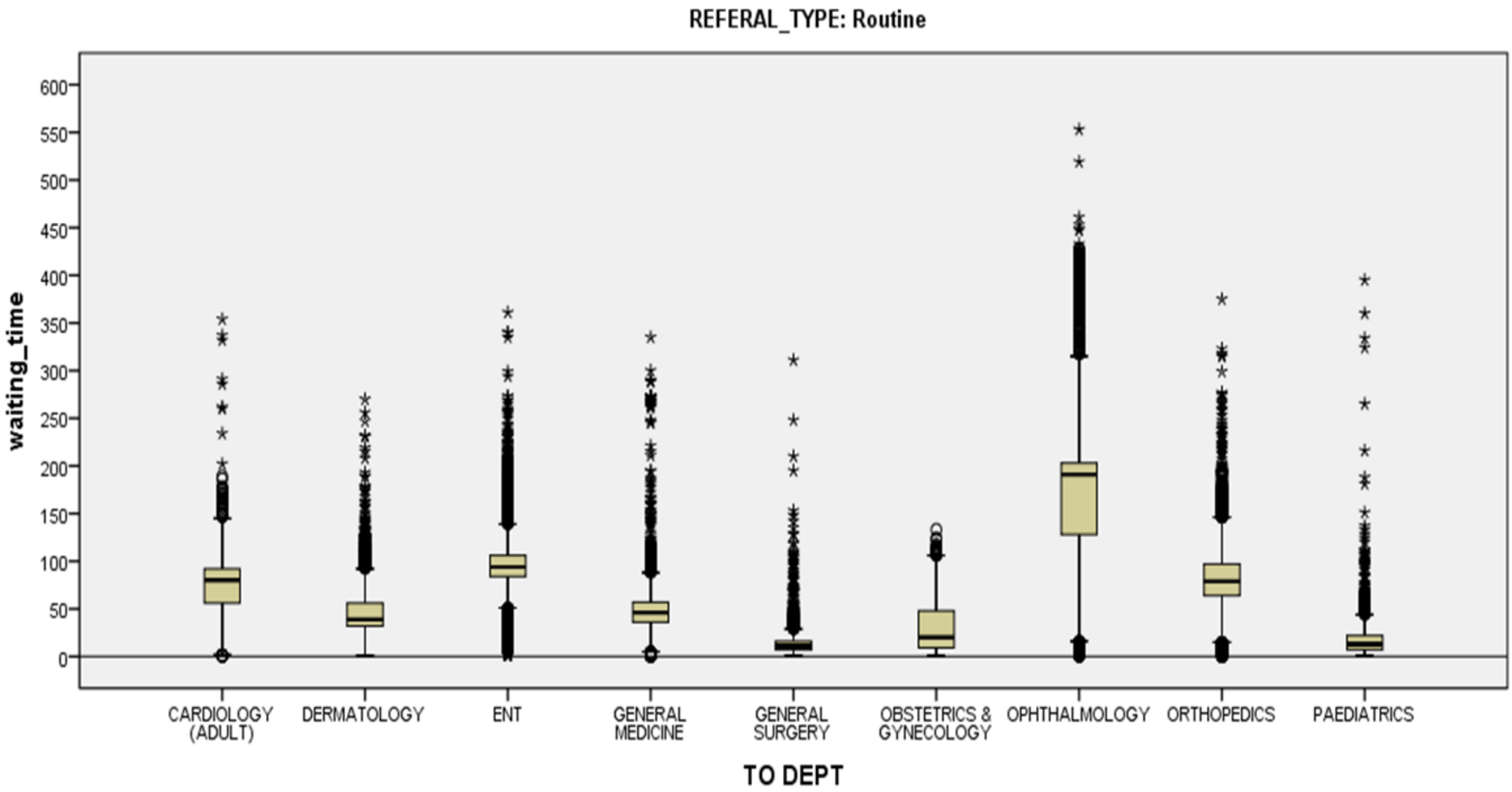
This indicates inconsistent waiting times, where some patients wait much longer than others.

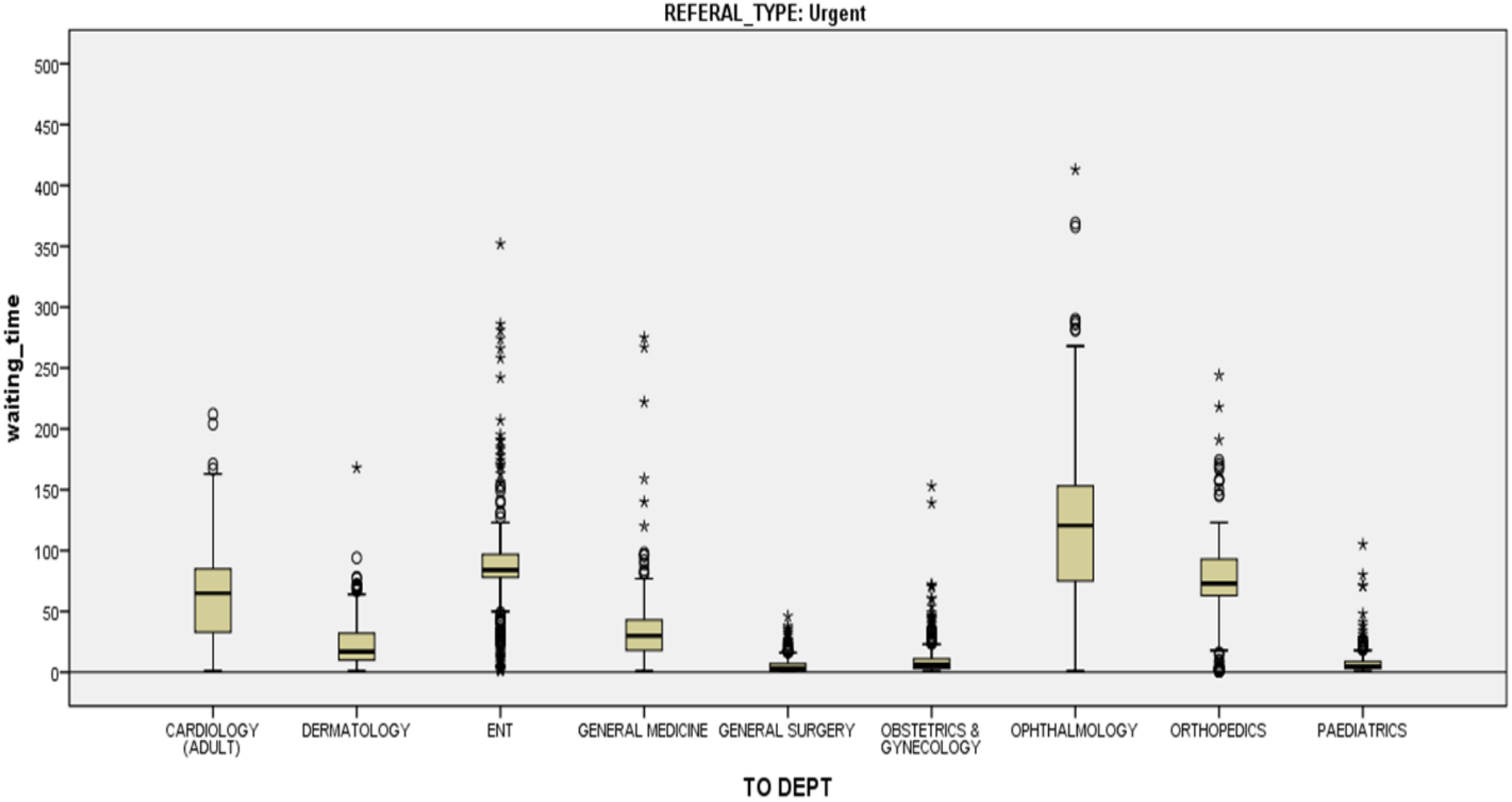
Outliers

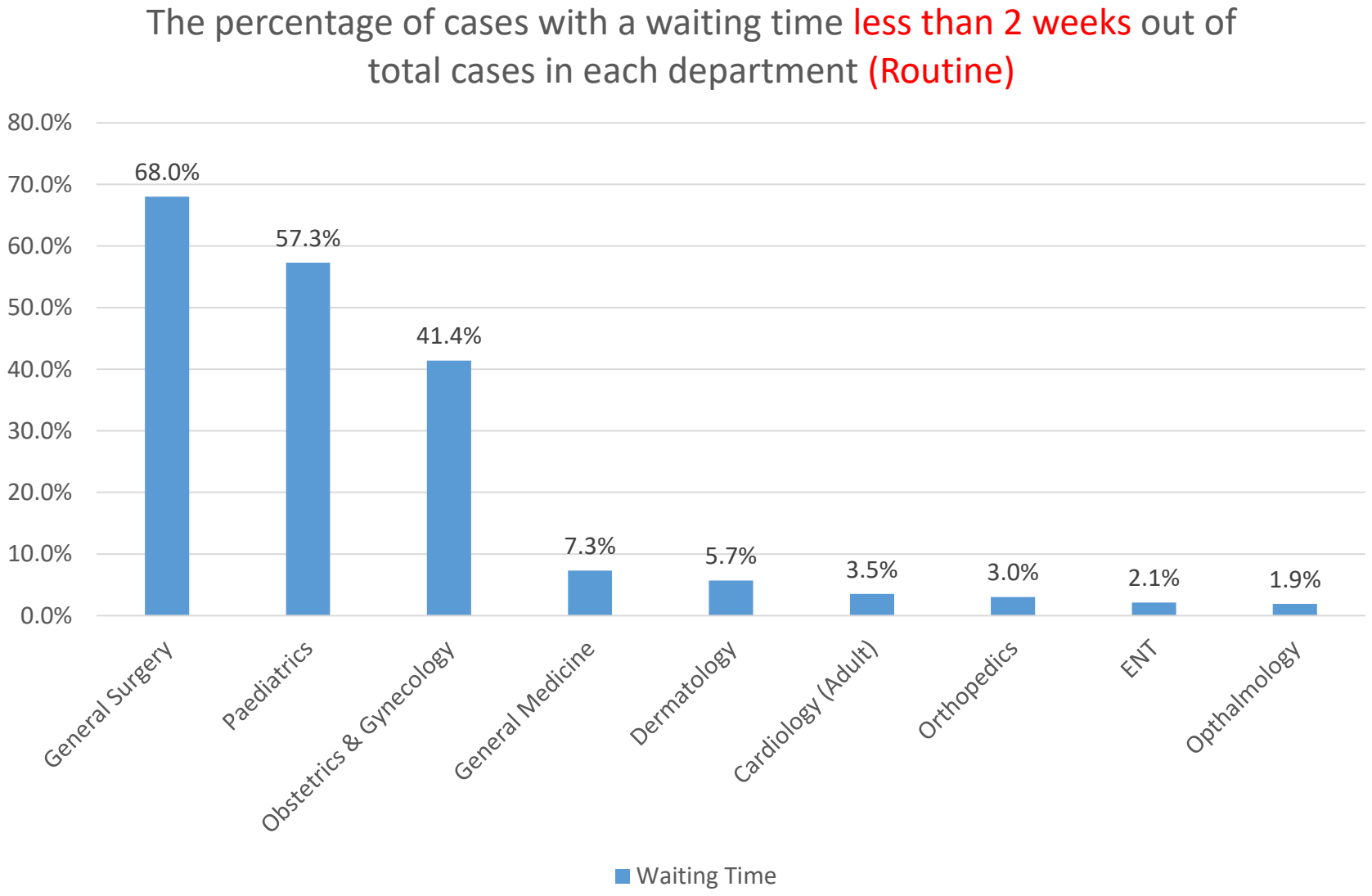
Many extreme values (stars/points) appear across departments, meaning:

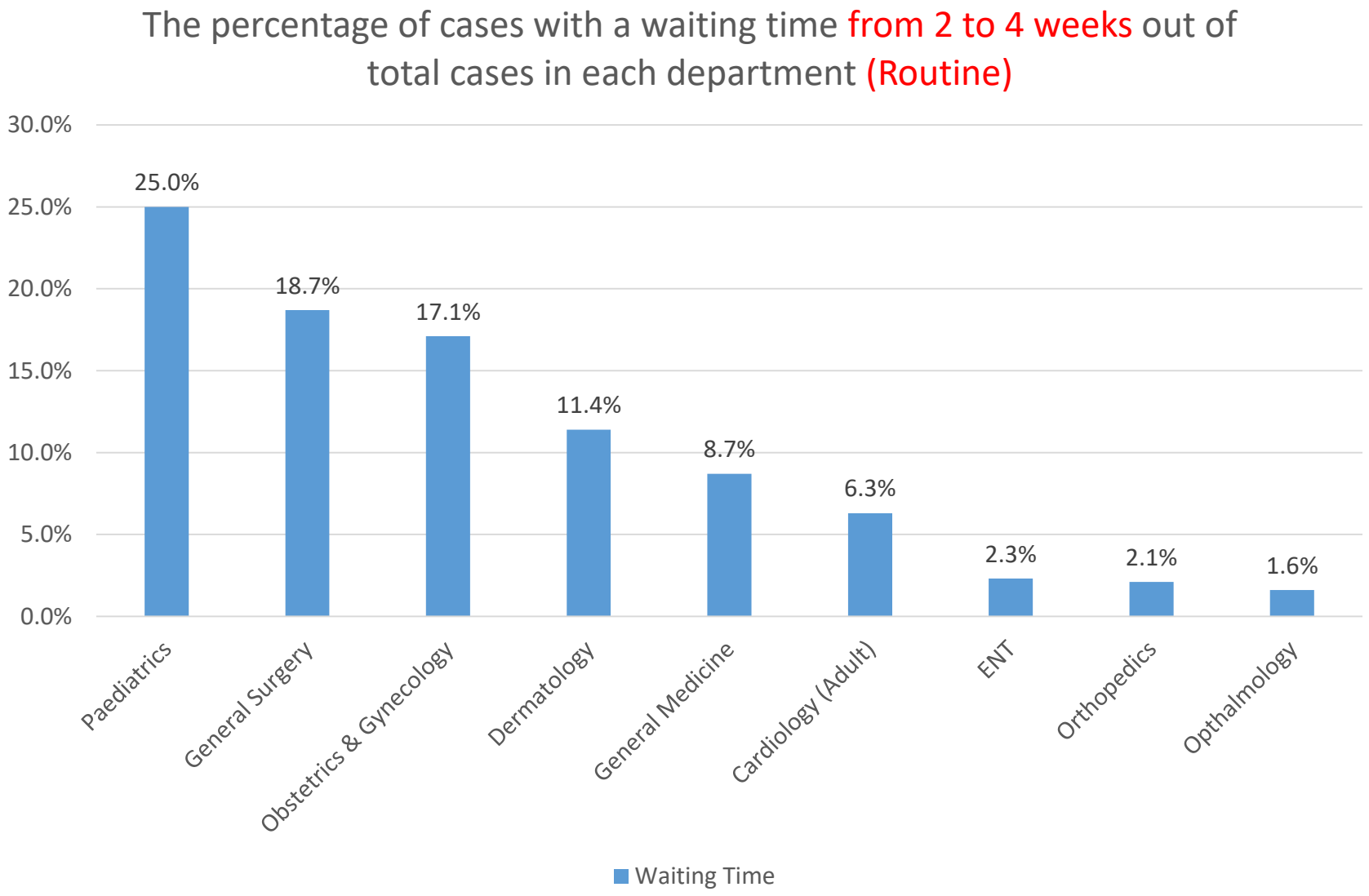
A number of patients experienced very long waits compared with the majority.

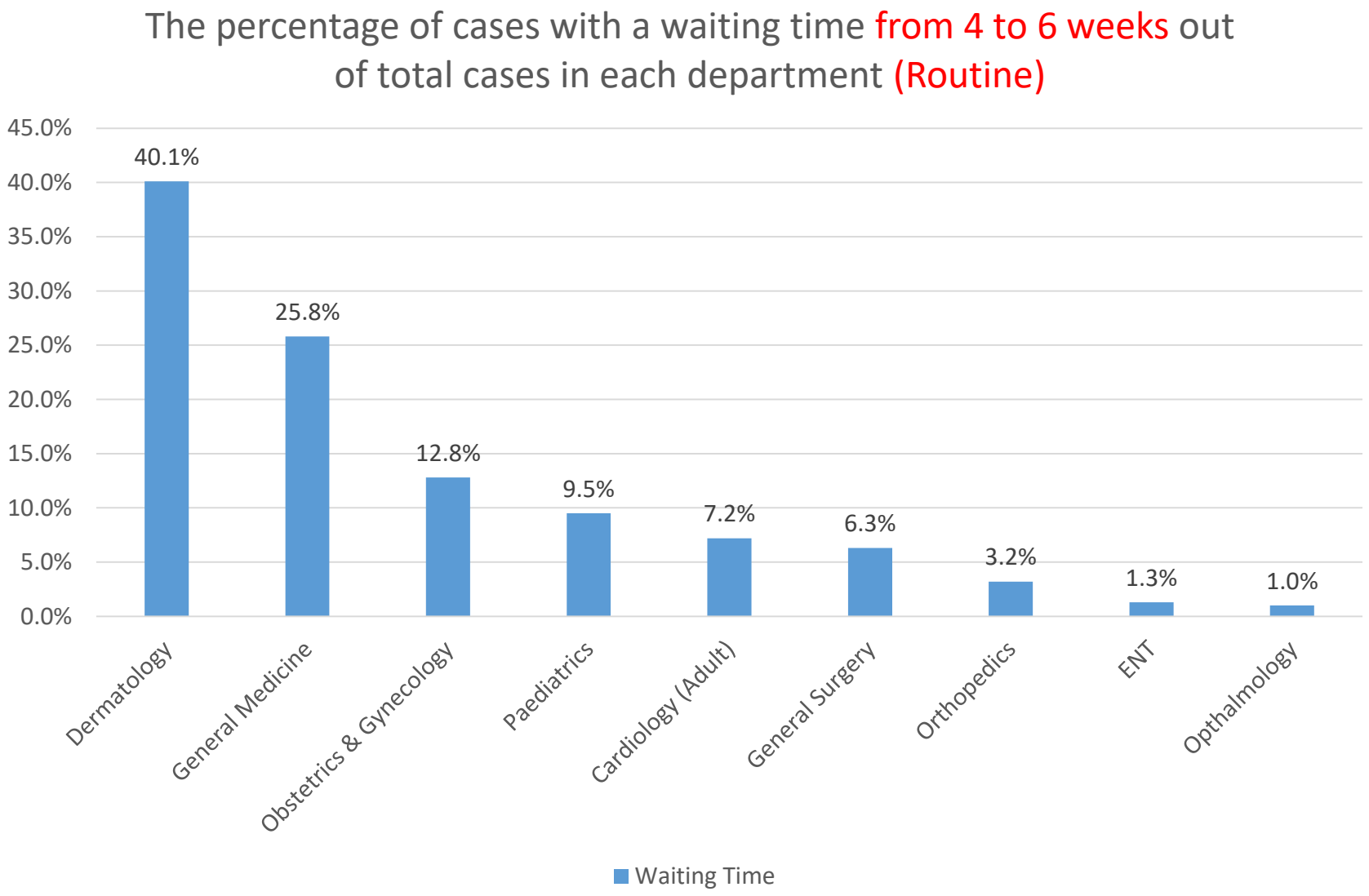
This could indicate system bottlenecks, backlog, or scheduling delays.

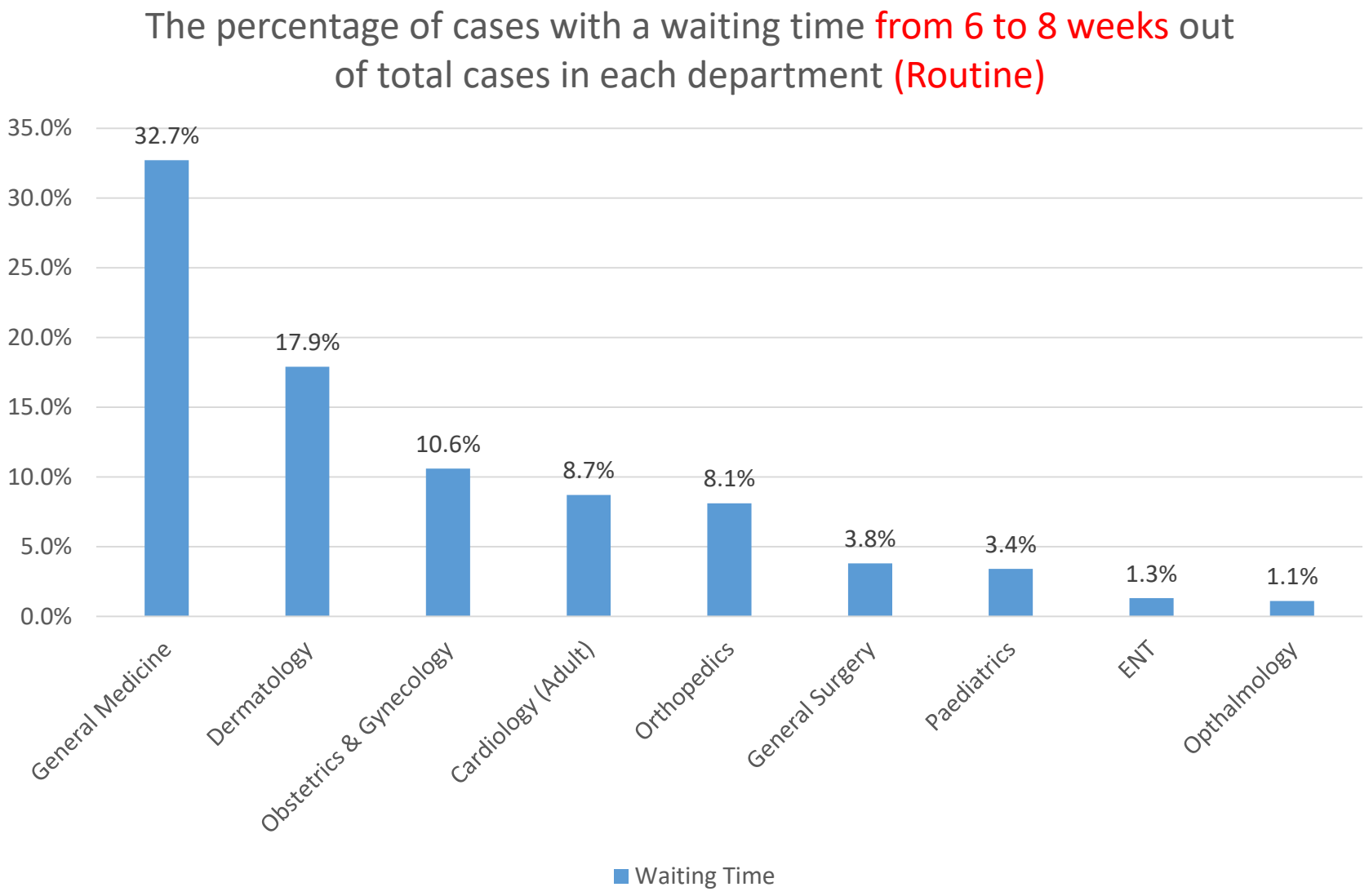

REFERAL_TYPE: Routine
waiting_time
600
550
500
450
400
350
300
250
200
150
100
50
0
CARDIOLOGY (ADULT)
DERMATOLOGY
ENT
GENERAL MEDICINE
GENERAL SURGERY
OBSTETRICS & GYNECOLOGY
OPHTHALMOLOGY
ORTHOPEDICS
PAEDIATRICS
TO DEPT

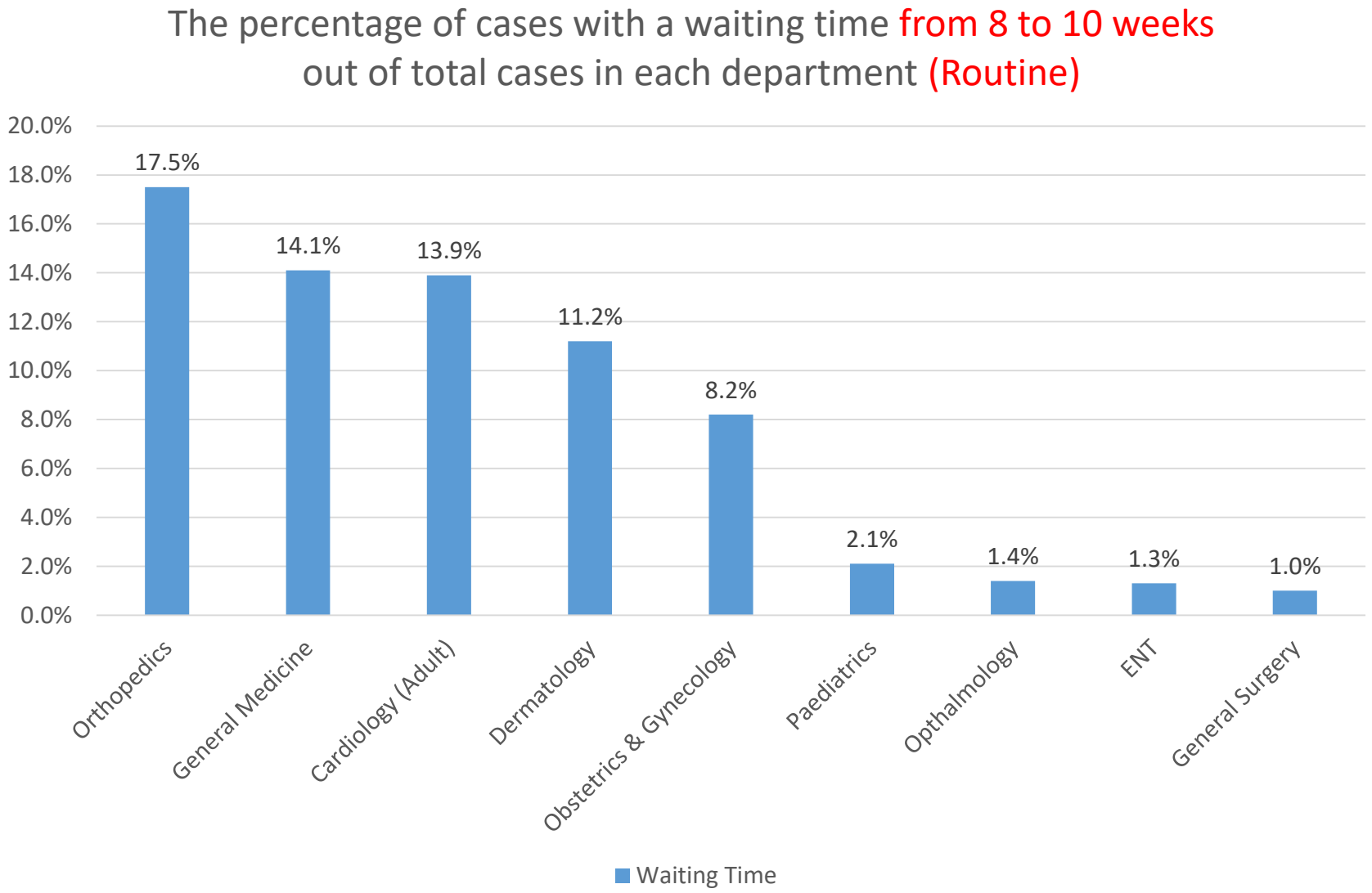

REFERAL_TYPE: Urgent
waiting_time
500
450
400
350
300
250
200
150
100
50
0
CARDIOLOGY (ADULT)
DERMATOLOGY
ENT
GENERAL MEDICINE
GENERAL SURGERY
OBSTETRICS & GYNECOLOGY
OPHTHALMOLOGY
ORTHOPEDICS
PAEDIATRICS
TO DEPT


The percentage of cases with a waiting time less than 2 weeks out of total cases in each department (Routine)
80.0%
70.0%
60.0%
50.0%
40.0%
30.0%
20.0%
10.0%
0.0%
68.0%
57.3%
41.4%
7.3%
5.7%
3.5%
3.0%
2.1%
1.9%
General Surgery
Paediatrics
Obstetrics & Gynecology
General Medicine
Dermatology
Cardiology (Adult)
Orthopedics
ENT
Opthalmology
Waiting Time

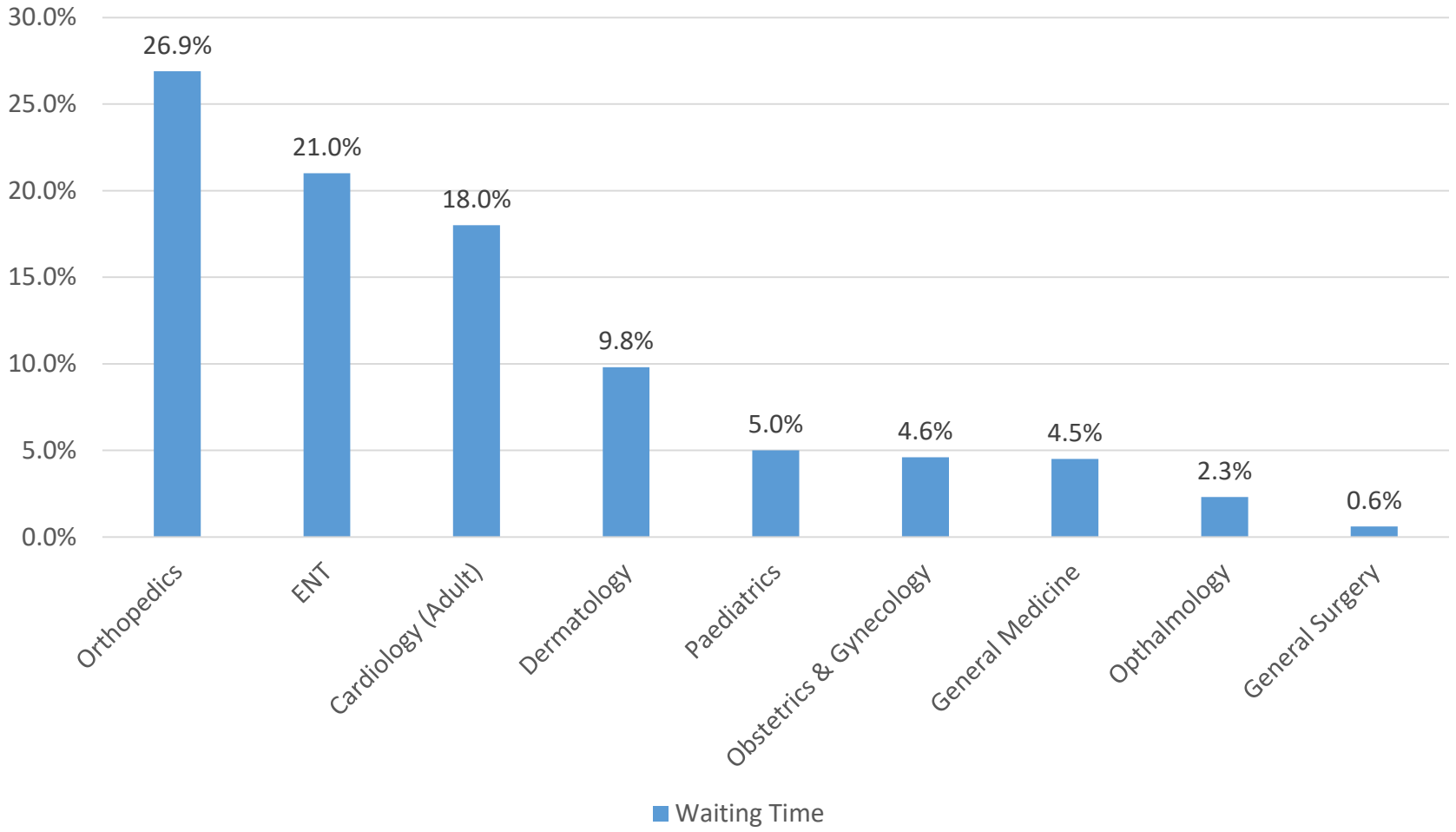

The percentage of cases with a waiting time from 2 to 4 weeks out of total cases in each department (Routine)
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
25.0%
18.7%
17.1%
11.4%
8.7%
6.3%
2.3%
2.1%
1.6%
Paediatrics
General Surgery
Obstetrics & Gynecology
Dermatology
General Medicine
Cardiology (Adult)
ENT
Orthopedics
Opthalmology
Waiting Time


The percentage of cases with a waiting time from 4 to 6 weeks out of total cases in each department (Routine)
45.0%
40.0%
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
40.1%
25.8%
12.8%
9.5%
7.2%
6.3%
3.2%
1.3%
1.0%
Dermatology
General Medicine
Obstetrics & Gynecology
Paediatrics
Cardiology (Adult)
General Surgery
Orthopedics
ENT
Opthalmology
Waiting Time

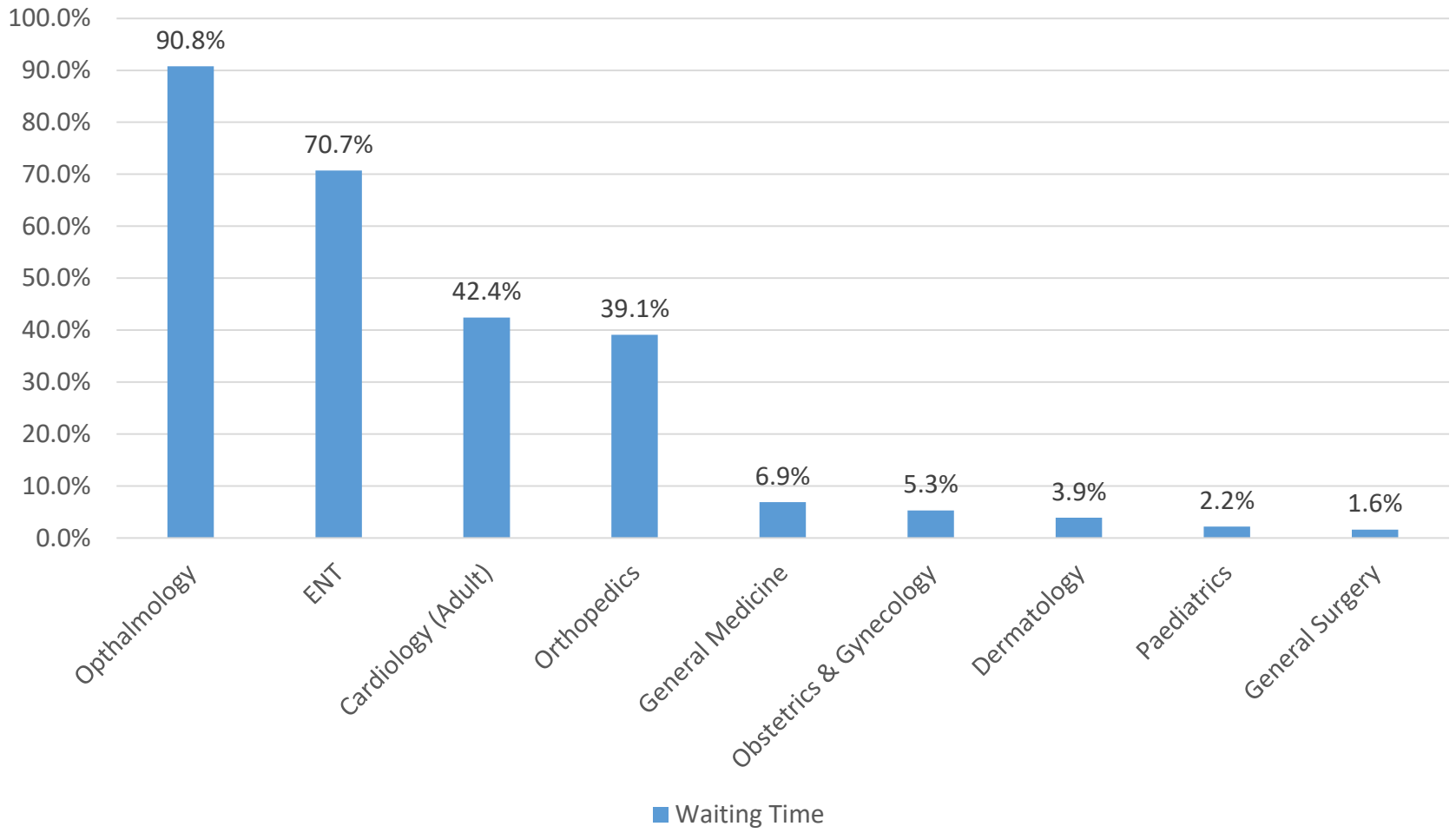

The percentage of cases with a waiting time from 6 to 8 weeks out of total cases in each department (Routine)
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
32.7%
17.9%
10.6%
8.7%
8.1%
3.8%
3.4%
1.3%
1.1%
General Medicine
Dermatology
Obstetrics & Gynecology
Cardiology (Adult)
Orthopedics
General Surgery
Paediatrics
ENT
Opthalmology
Waiting Time

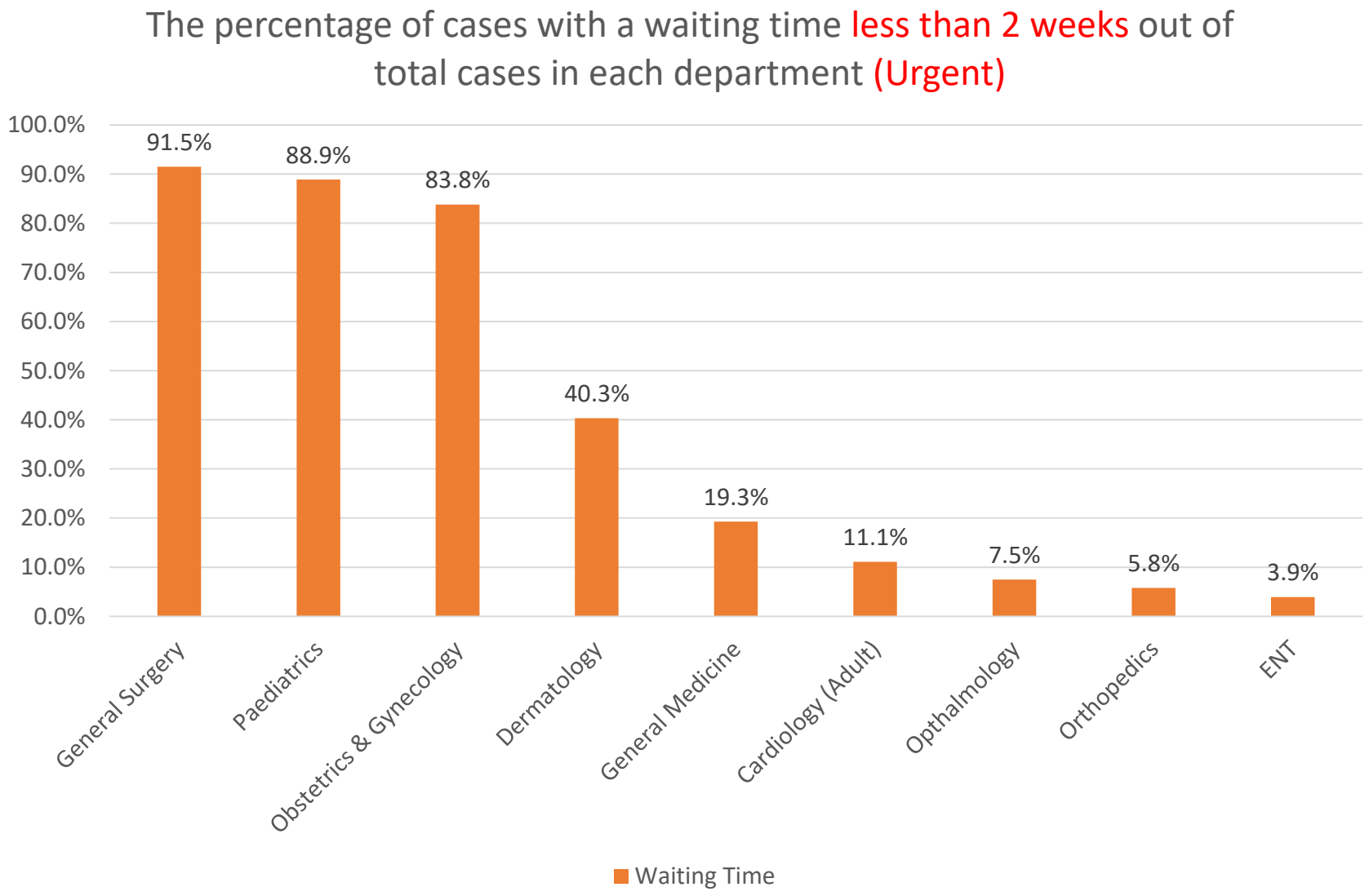

The percentage of cases with a waiting time from 8 to 10 weeks
out of total cases in each department (Routine)
20.0%
18.0%
16.0%
14.0%
12.0%
10.0%
8.0%
6.0%
4.0%
2.0%
0.0%
17.5%
14.1%
13.9%
11.2%
8.2%
2.1%
1.4%
1.3%
1.0%
Orthopedics
General Medicine
Cardiology (Adult)
Dermatology
Obstetrics & Gynecology
Paediatrics
Opthalmology
ENT
General Surgery
Waiting Time

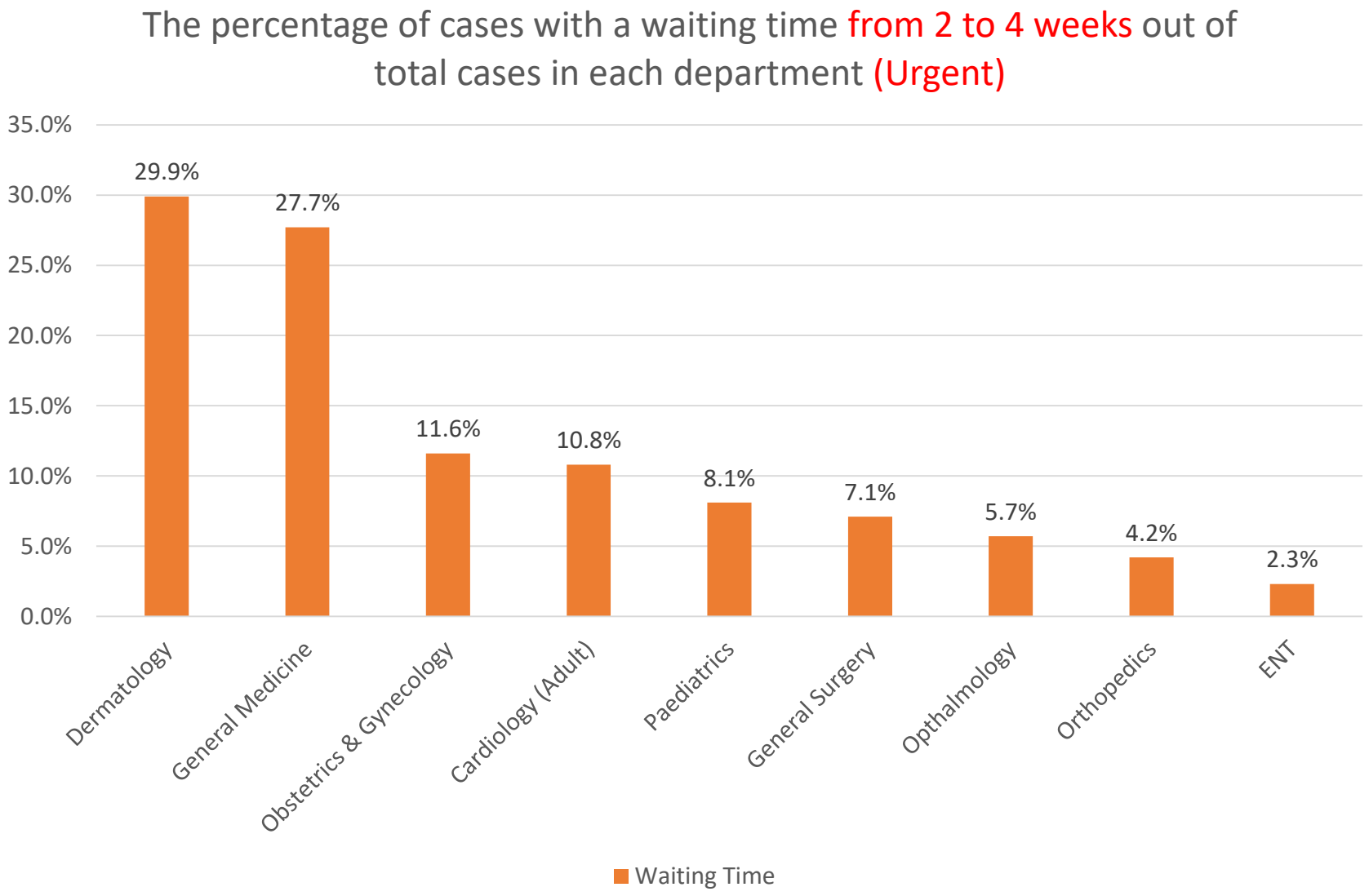
The percentage of cases with a waiting time from 10 to 12 weeks out of total cases in each department (Routine)

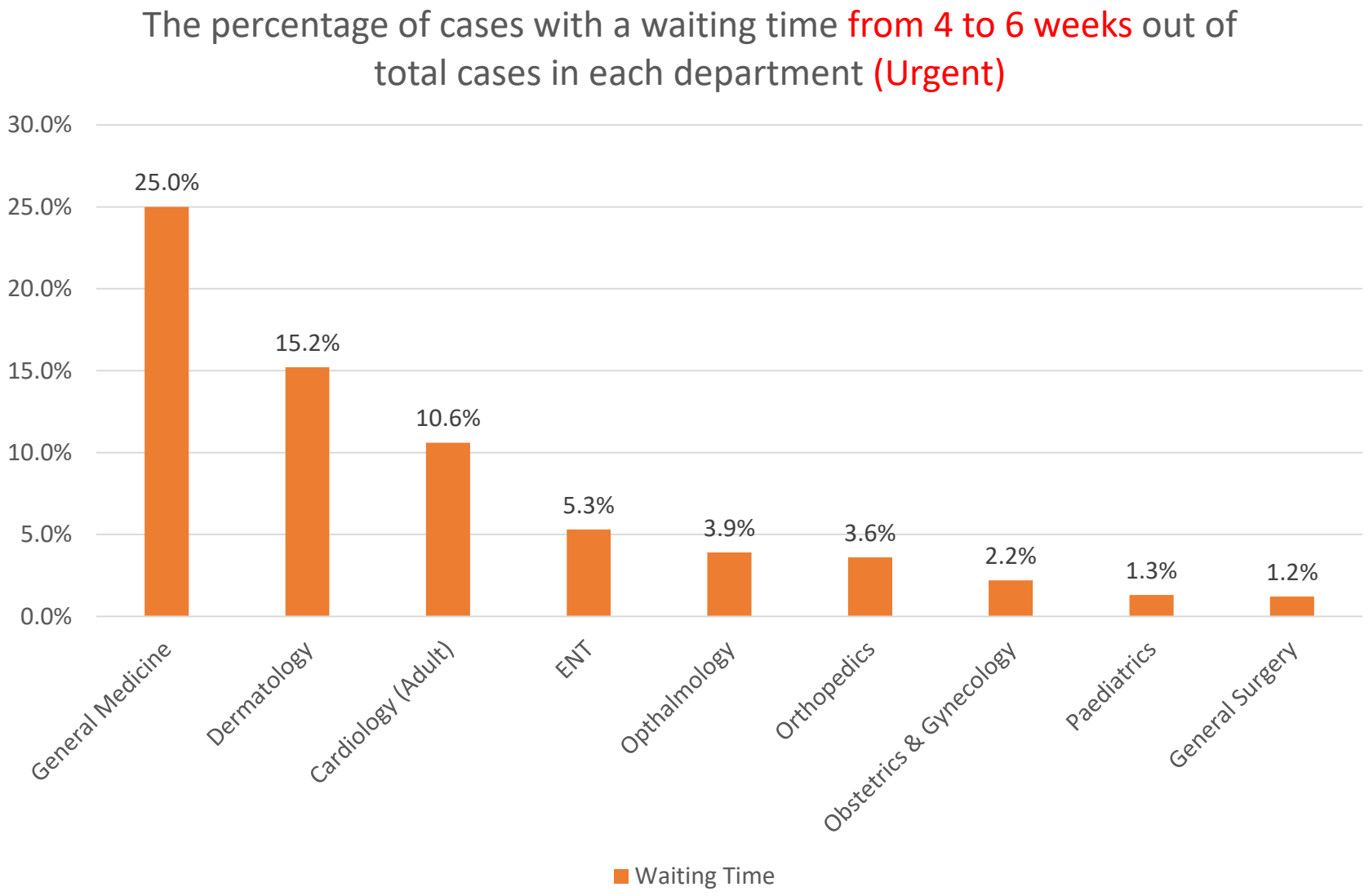


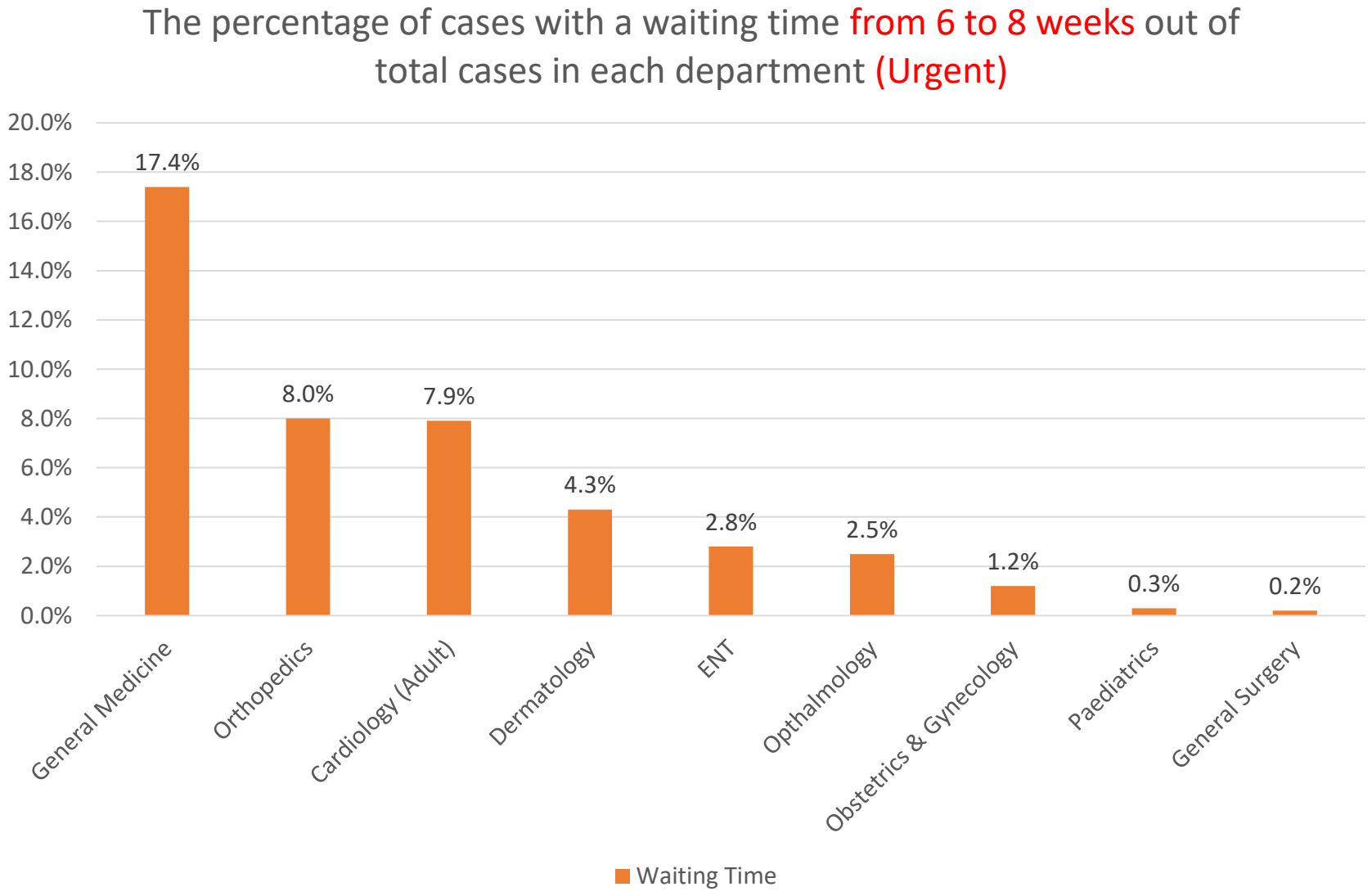
The percentage of cases with a waiting time more than 12 weeks out of total cases in each department (Routine)

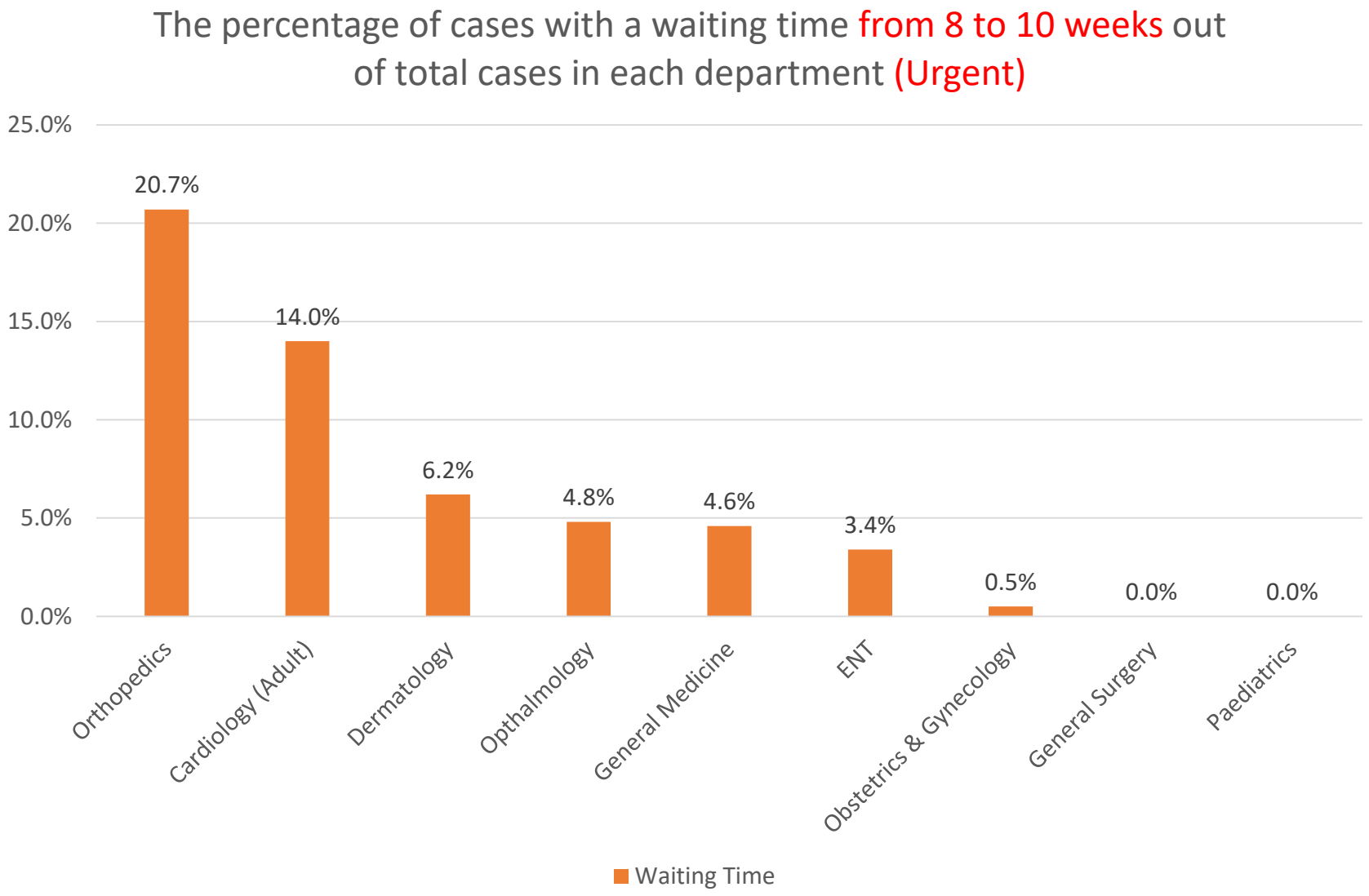


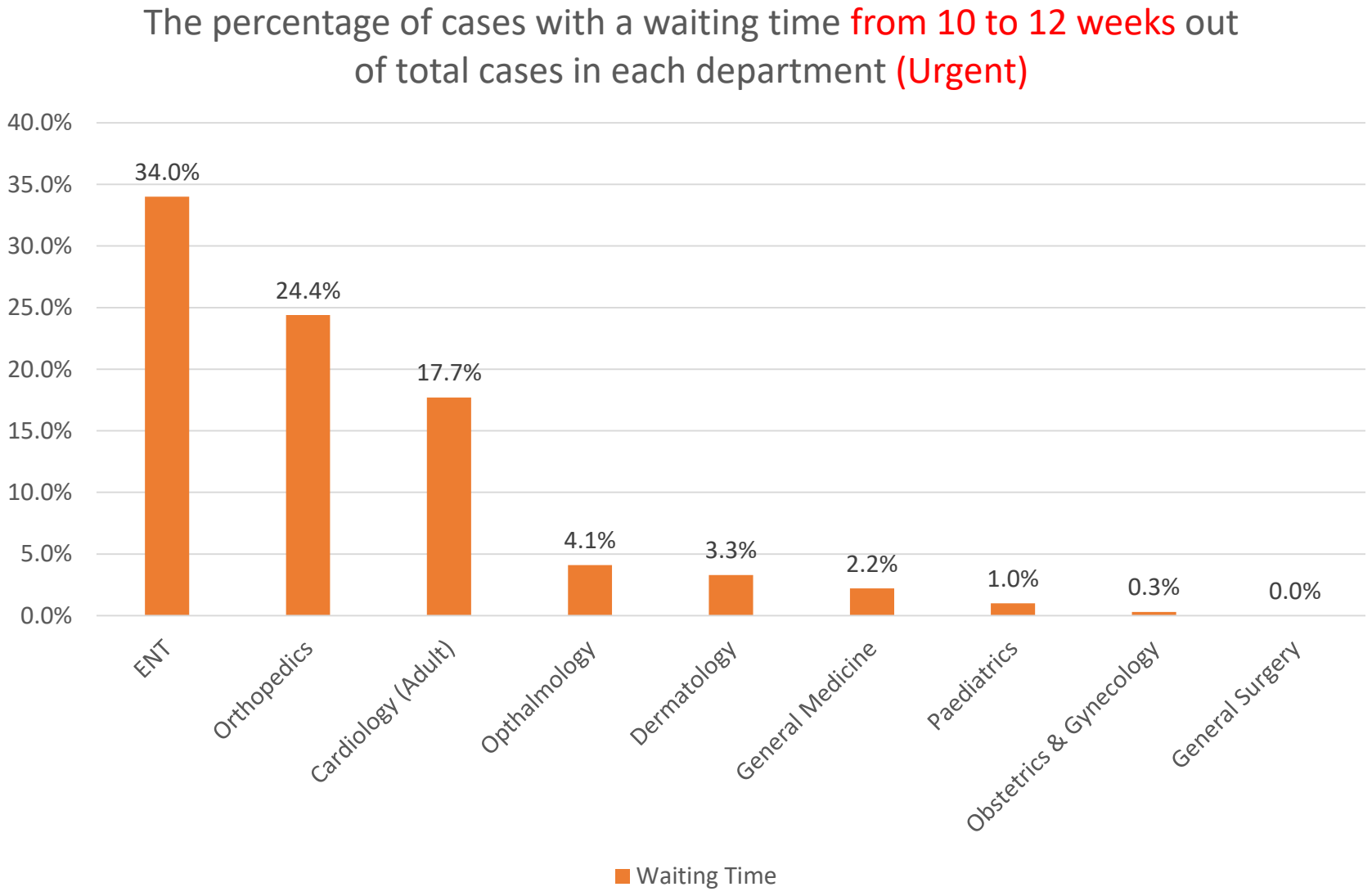

The percentage of cases with a waiting time less than 2 weeks out of total cases in each department (Urgent)
100.0%
90.0%
80.0%
70.0%
60.0%
50.0%
40.0%
30.0%
20.0%
10.0%
0.0%
91.5%
88.9%
83.8%
40.3%
19.3%
11.1%
7.5%
5.8%
3.9%
General Surgery
Paediatrics
Obstetrics & Gynecology
Dermatology
General Medicine
Cardiology (Adult)
Opthalmology
Orthopedics
ENT
Waiting Time

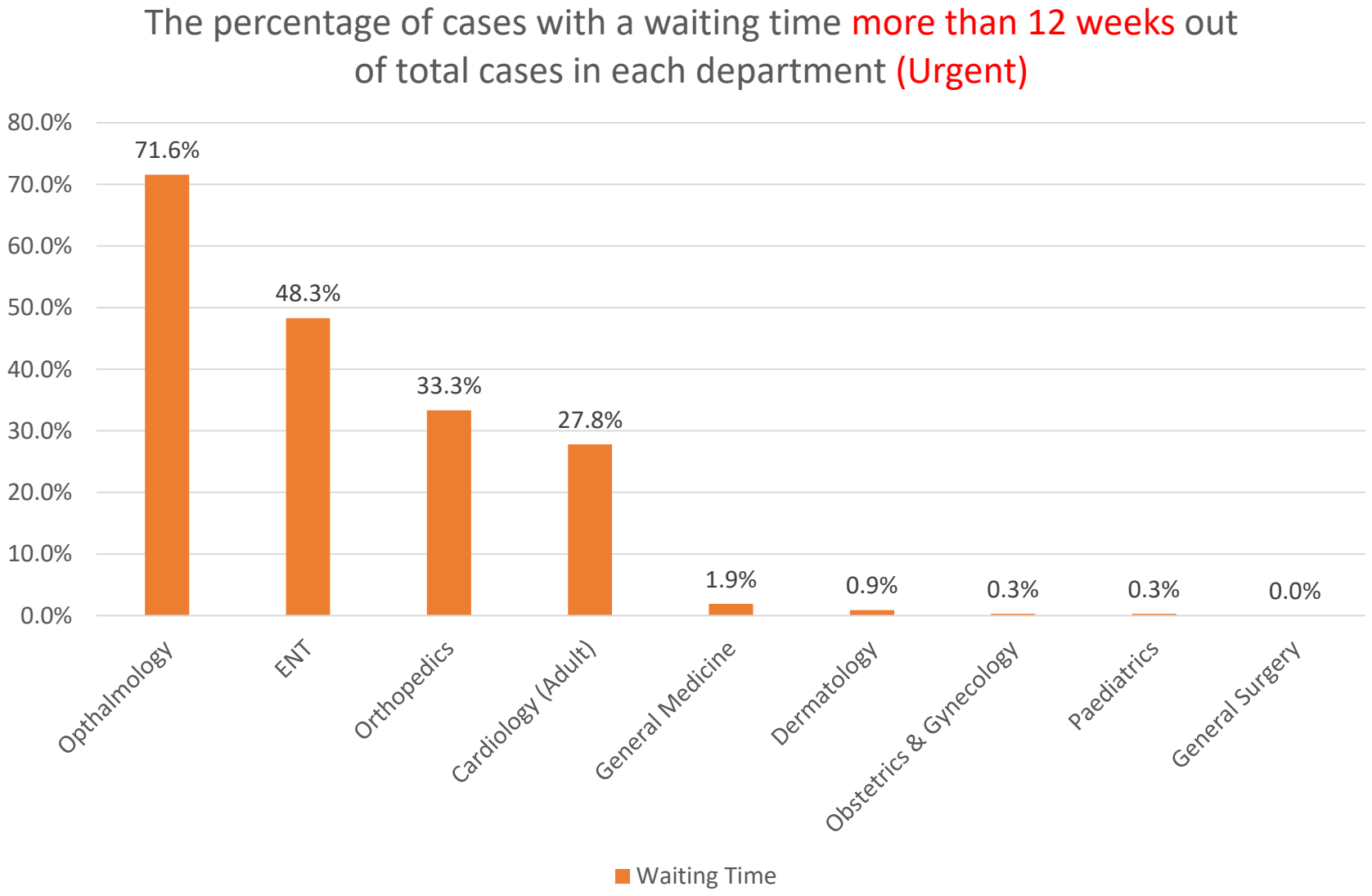

The percentage of cases with a waiting time from 2 to 4 weeks out of total cases in each department (Urgent)
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
29.9%
27.7%
11.6%
10.8%
8.1%
7.1%
5.7%
4.2%
2.3%
Dermatology
General Medicine
Obstetrics & Gynecology
Cardiology (Adult)
Paediatrics
General Surgery
Opthalmology
Orthopedics
ENT
Waiting Time


The percentage of cases with a waiting time from 4 to 6 weeks out of total cases in each department (Urgent)
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
25.0%
15.2%
10.6%
5.3%
3.9%
3.6%
2.2%
1.3%
1.2%
General Medicine
Dermatology
Cardiology (Adult)
ENT
Opthalmology
Orthopedics
Obstetrics & Gynecology
Paediatrics
General Surgery
Waiting Time


The percentage of cases with a waiting time from 6 to 8 weeks out of total cases in each department (Urgent)
20.0%
18.0%
16.0%
14.0%
12.0%
10.0%
8.0%
6.0%
4.0%
2.0%
0.0%
17.4%
8.0%
7.9%
4.3%
2.8%
2.5%
1.2%
0.3%
0.2%
General Medicine
Orthopedics
Cardiology (Adult)
Dermatology
ENT
Opthalmology
Obstetrics & Gynecology
Paediatrics
General Surgery
Waiting Time


The percentage of cases with a waiting time from 8 to 10 weeks out of total cases in each department (Urgent)
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
20.7%
14.0%
6.2%
4.8%
4.6%
3.4%
0.5%
0.0%
0.0%
Orthopedics
Cardiology (Adult)
Dermatology
Opthalmology
General Medicine
ENT
Obstetrics & Gynecology
General Surgery
Paediatrics
Waiting Time


The percentage of cases with a waiting time from 10 to 12 weeks out of total cases in each department (Urgent)
40.0%
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
34.0%
24.4%
17.7%
4.1%
3.3%
2.2%
1.0%
0.3%
0.0%
ENT
Orthopedics
Cardiology (Adult)
Opthalmology
Dermatology
General Medicine
Paediatrics
Obstetrics & Gynecology
General Surgery
Waiting Time

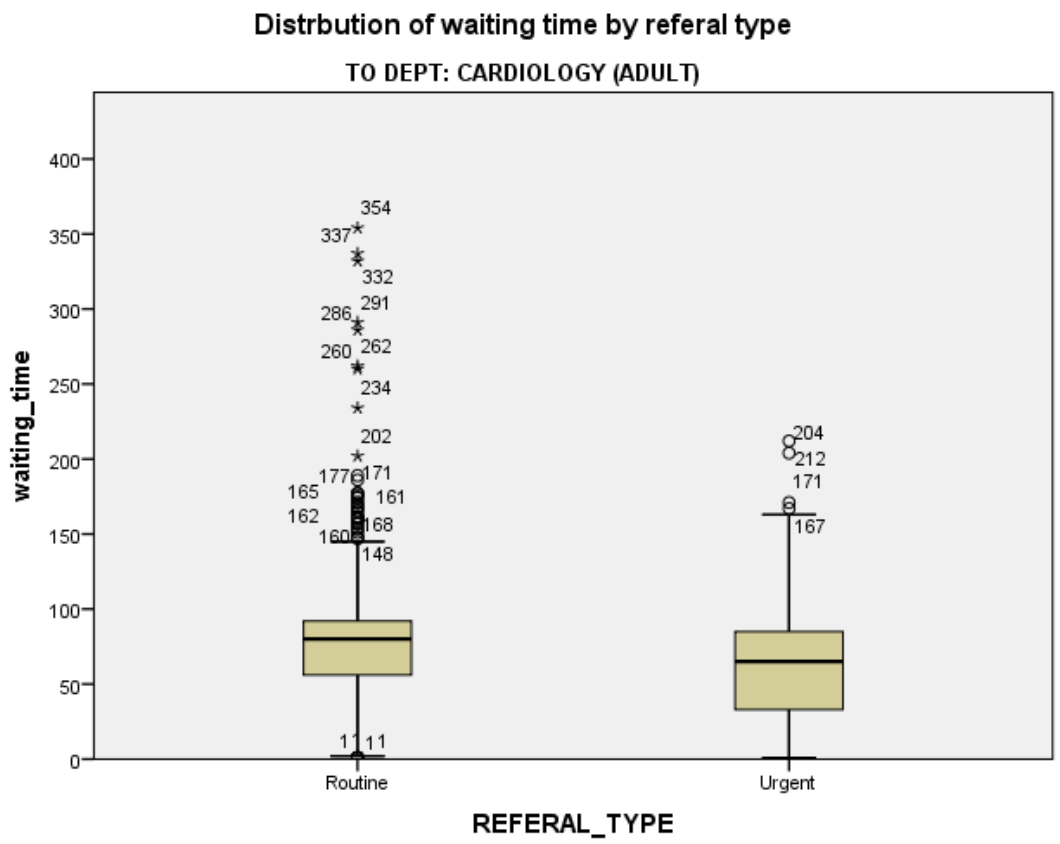

The percentage of cases with a waiting time more than 12 weeks out of total cases in each department (Urgent)
80.0%
70.0%
60.0%
50.0%
40.0%
30.0%
20.0%
10.0%
0.0%
71.6%
48.3%
33.3%
27.8%
1.9%
0.9%
0.3%
0.3%
0.0%
Opthalmology
ENT
Orthopedics
Cardiology (Adult)
General Medicine
Dermatology
Obstetrics & Gynecology
Paediatrics
General Surgery
Waiting Time

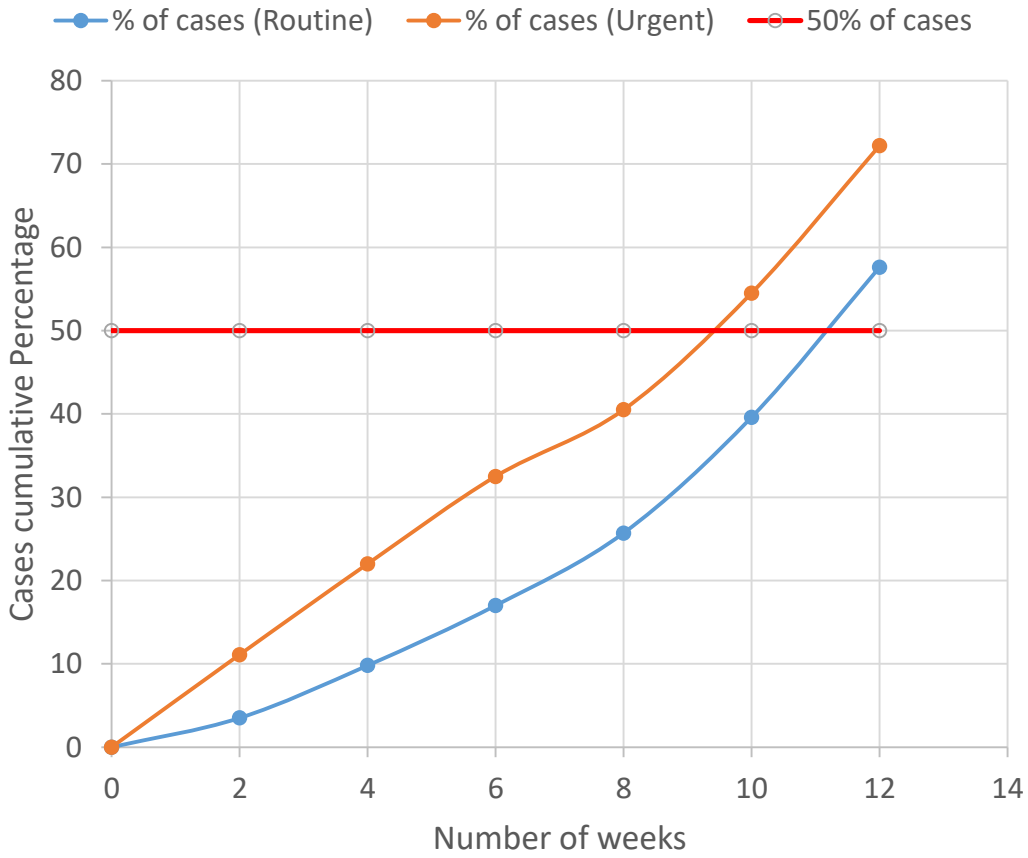
Section 2
waiting time by departments

Cardiology (Adult)

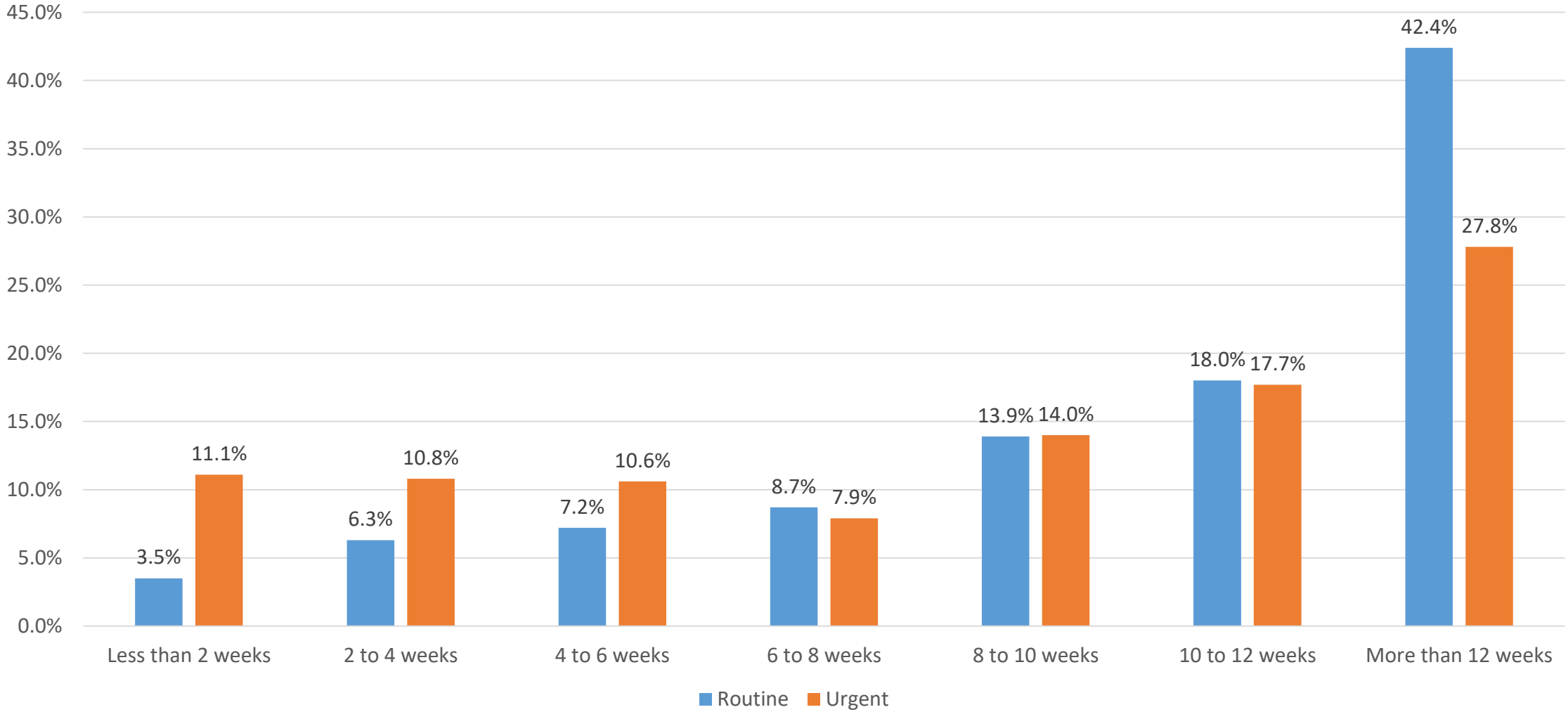
Routine: The median waiting time is approximately 80 days. There is a heavy cluster of outliers stretching up to 354 days. Additionally, 50% of total routine cases have a waiting time of more than 11 weeks.

Urgent: The median is slightly lower, around 65 days. While the spread is tighter than routine, some urgent cases still waited over 200 days. Furthermore, 50% of total urgent cases have a waiting time of more than 9 weeks.





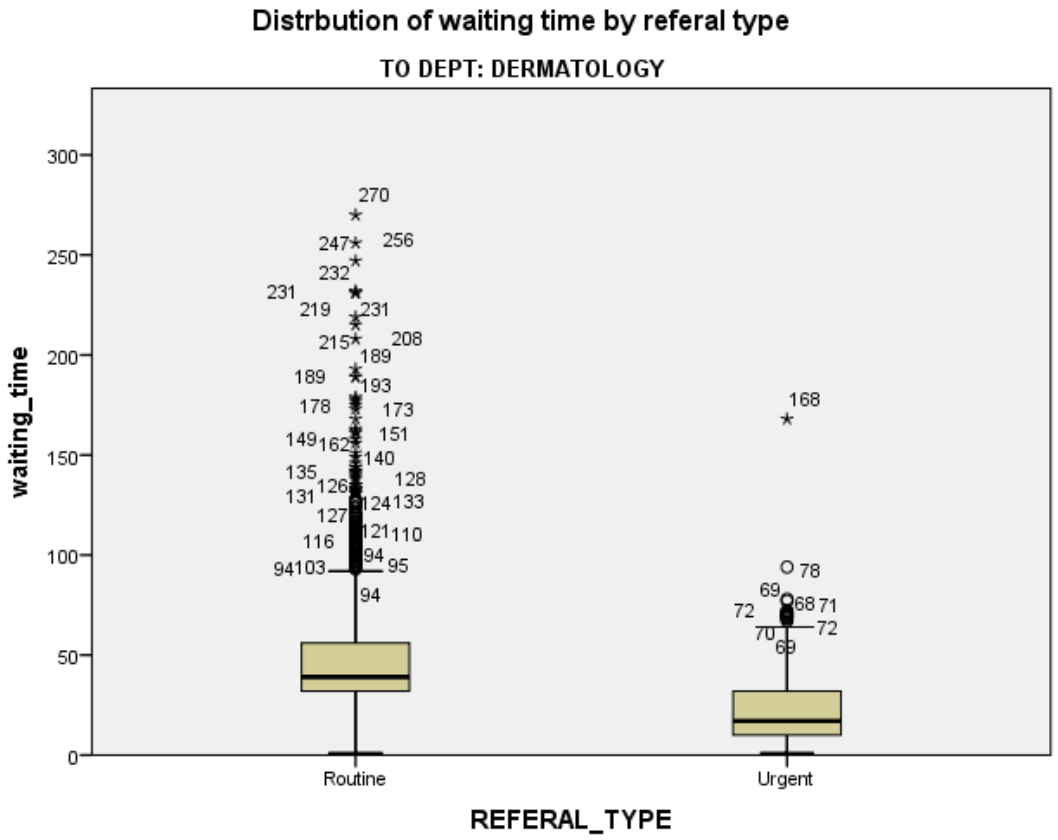
Waiting time by referral type (Cardiology)

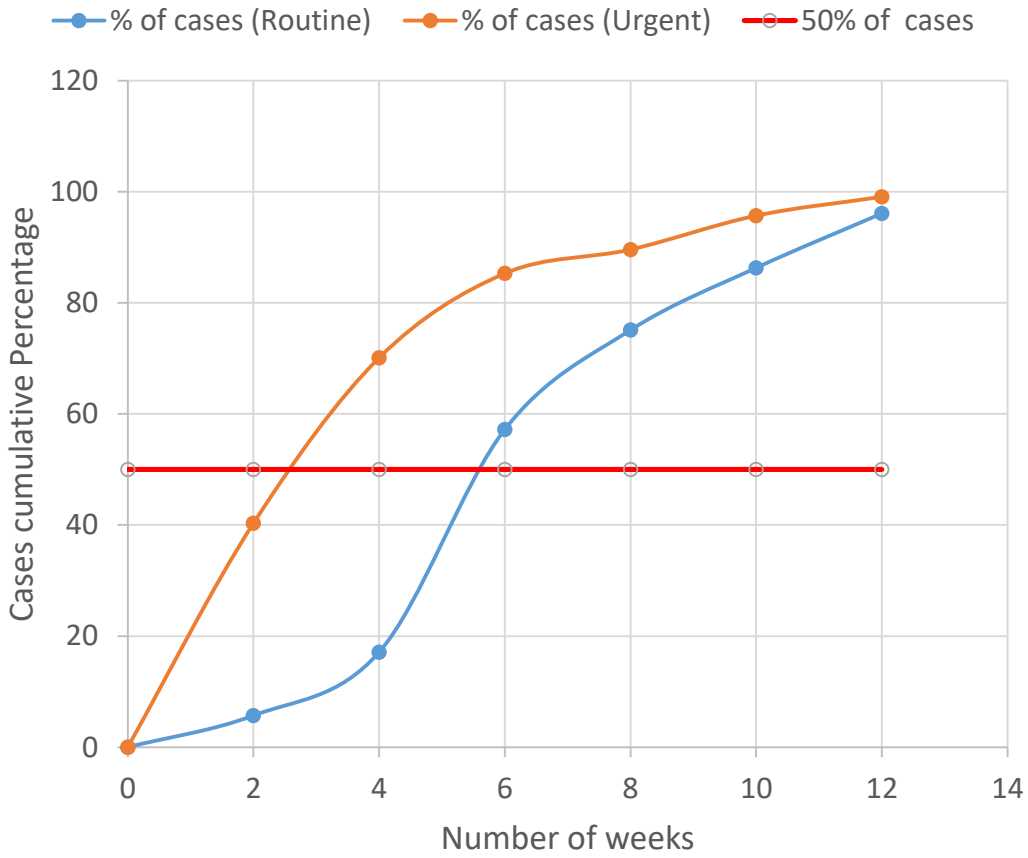


Dermatology

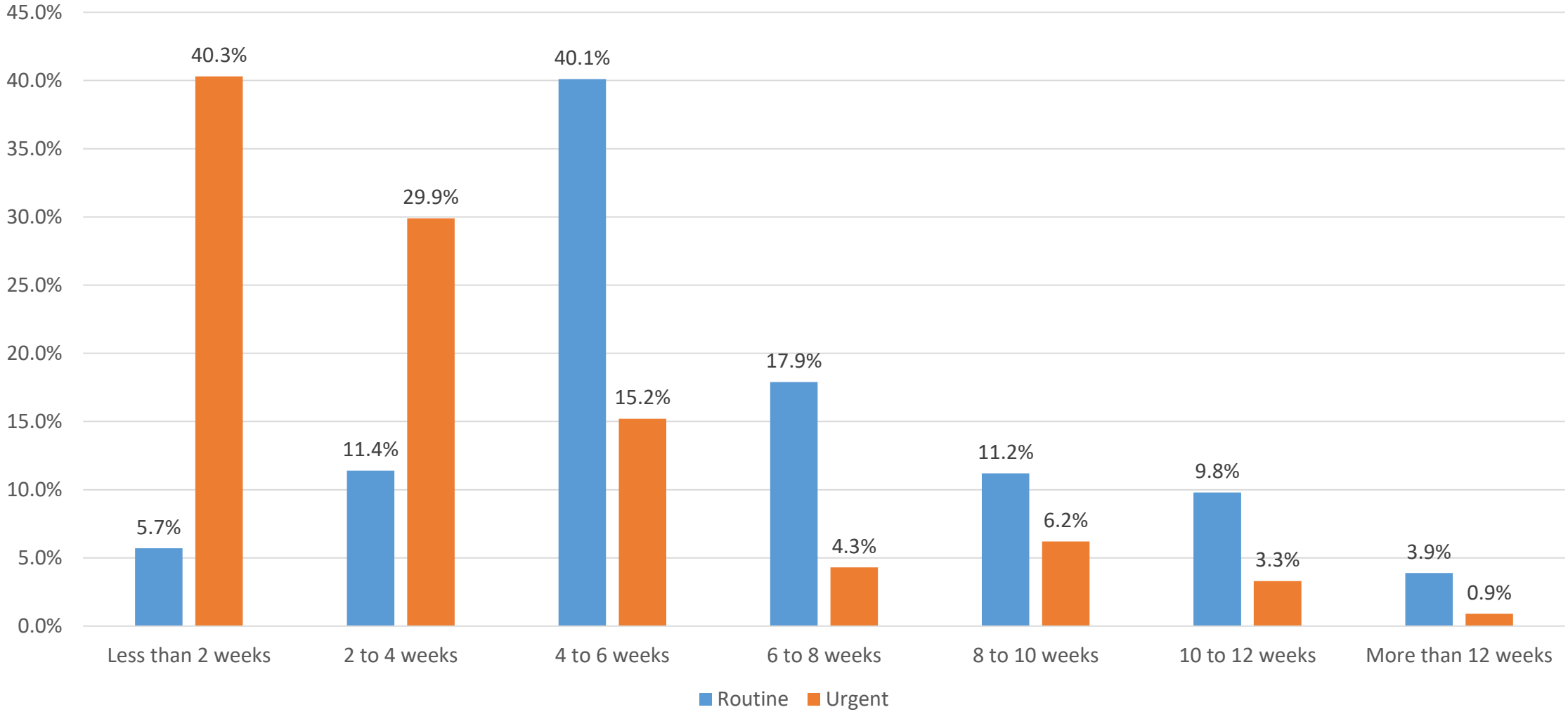
Routine: Patients typically wait around 40 days. However, there are extreme outliers reaching as high as 270 days. Additionally, 50% of total routine cases have a waiting time of more than 5 weeks.

Urgent: This department shows good prioritization, with a median wait time under 20 days. The maximum outlier for urgent cases is significantly lower than in Cardiology, at 168 days. Furthermore, 50% of total urgent cases have a waiting time of more than 2 weeks.





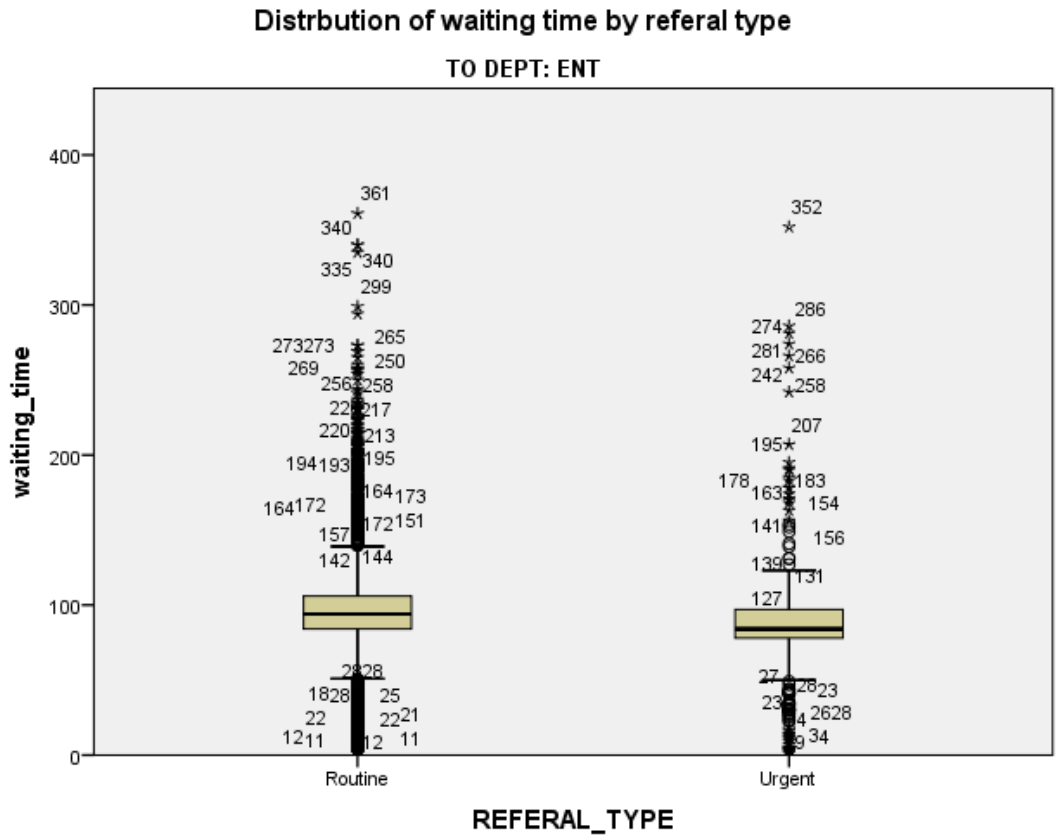
Waiting time by referral type (Dermatology)

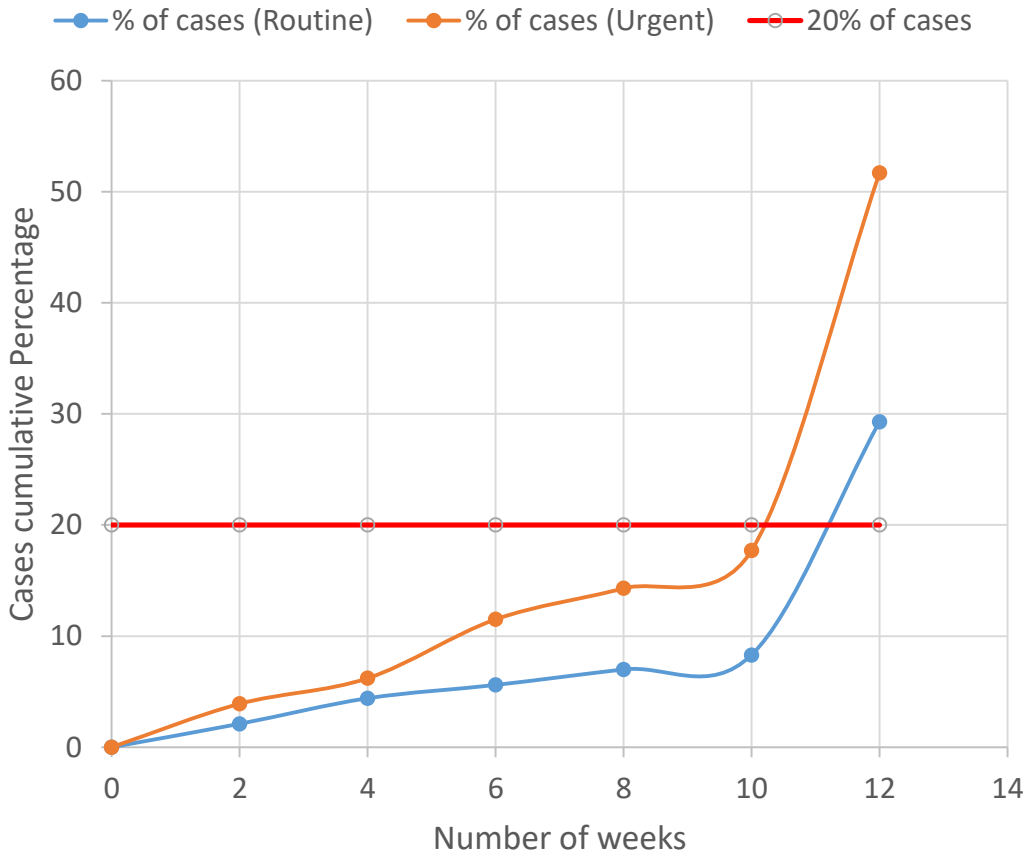


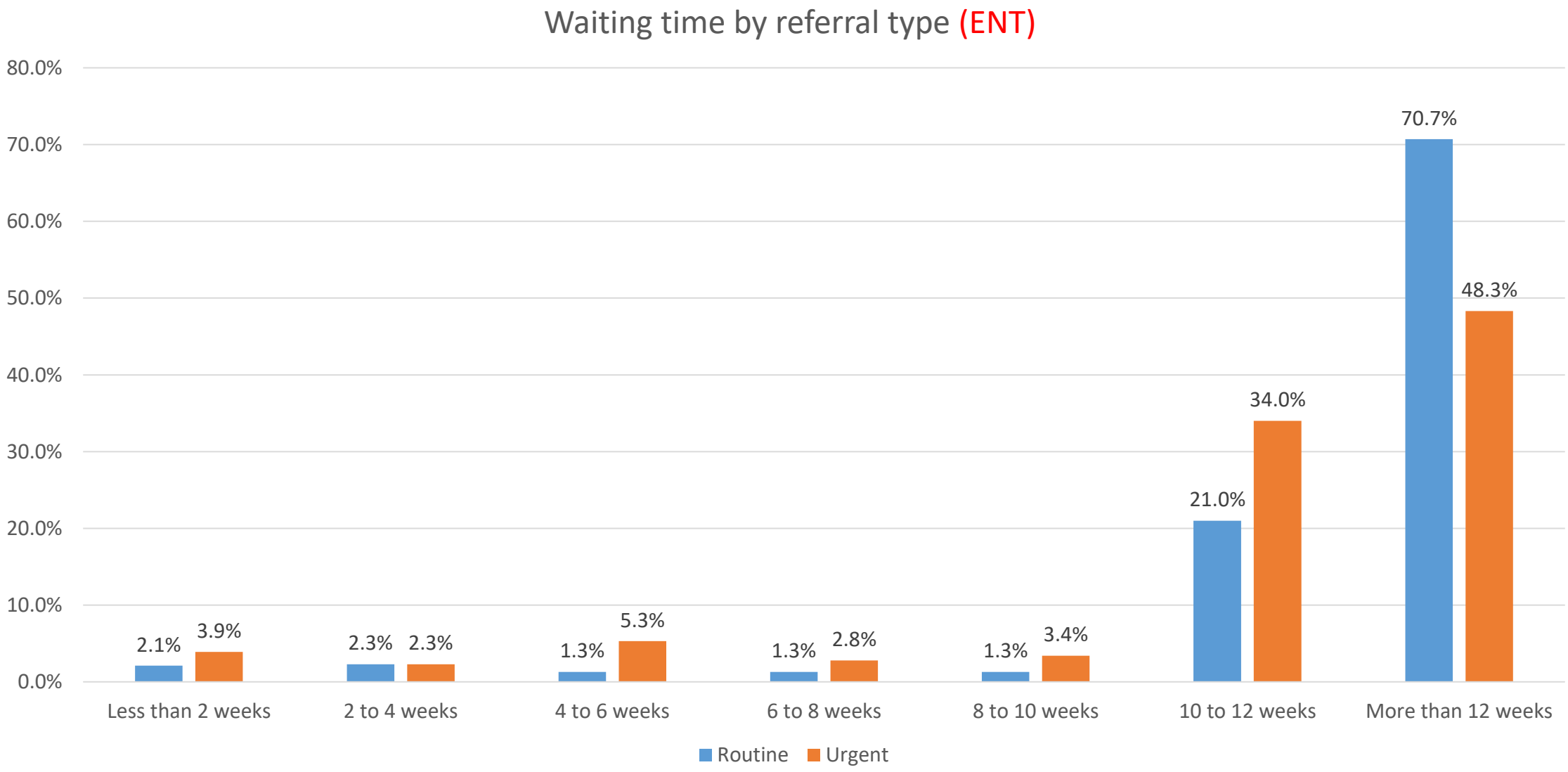
ENT

Routine: The median wait is roughly 90 days. The data is very "noisy" with a massive number of outliers starting from 142 days and peaking at 361 days. Moreover, 80% of total routine cases have a waiting time of more than 11 weeks.

Urgent: The median is nearly identical to routine (approx. 85 days), suggesting less differentiation between priority types in this department. Urgent outliers are also high, reaching 352 days. In addition, 80% of total urgent cases have a waiting time of more than 10 weeks.



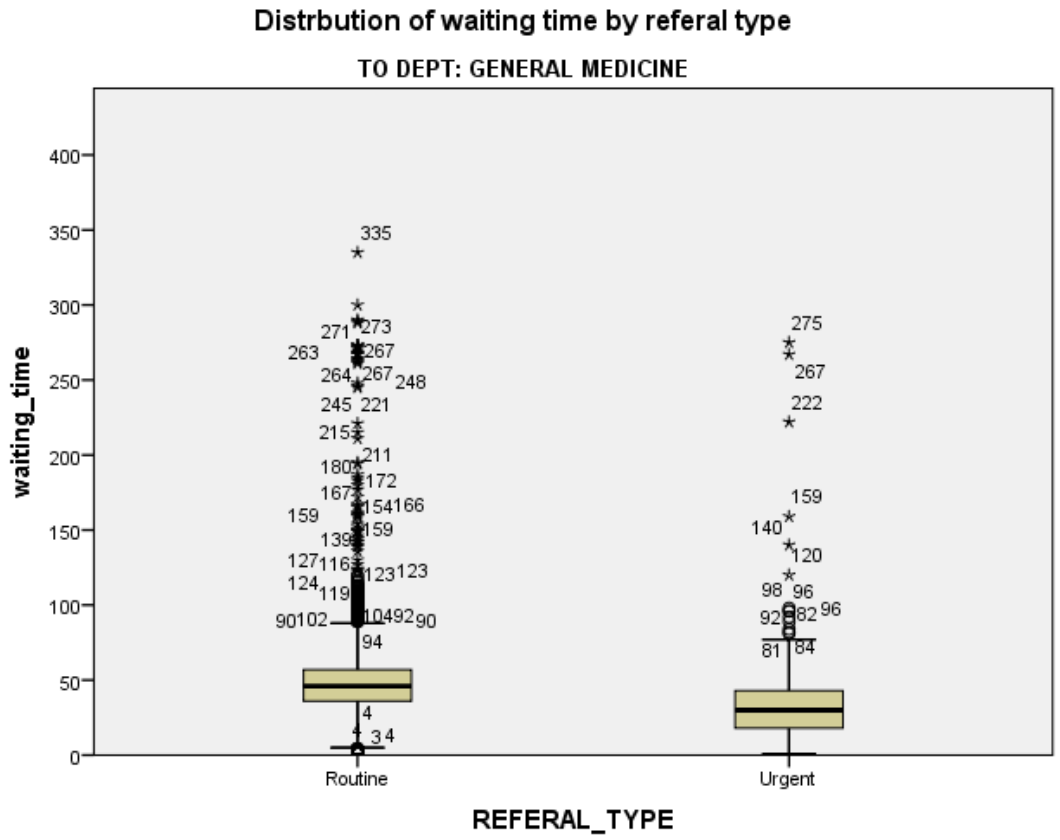


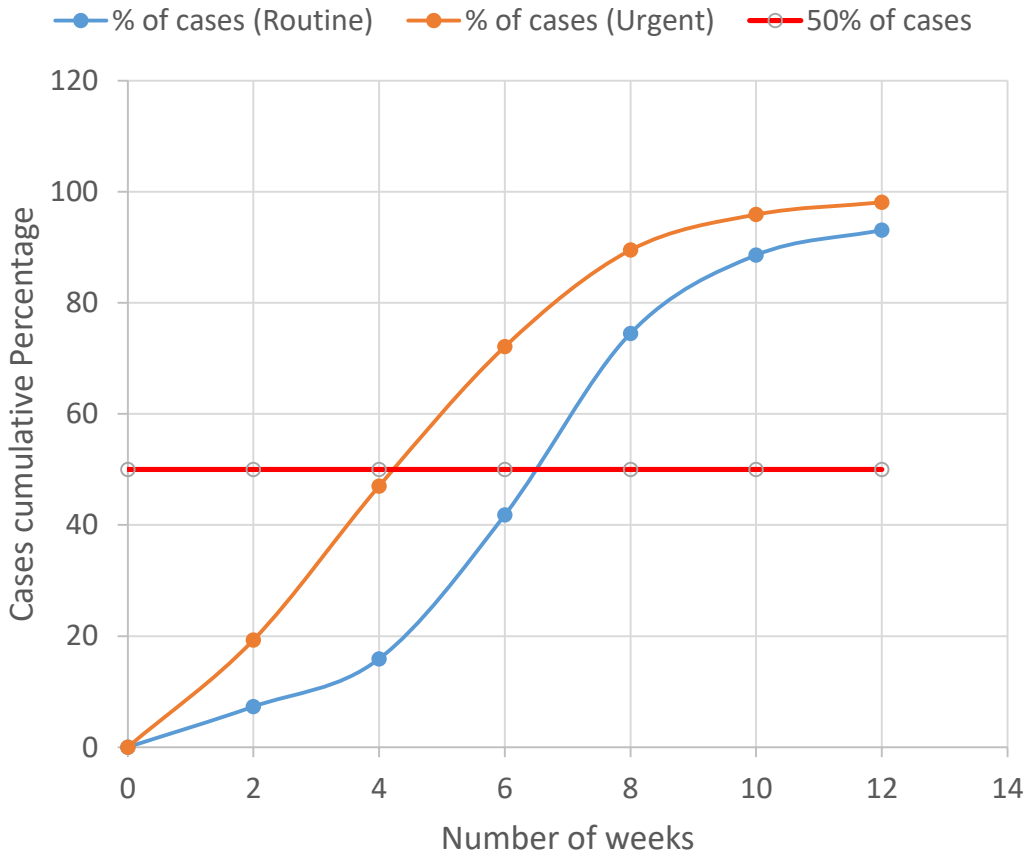

Waiting time by referral type (ENT)
80.0%
70.0%
60.0%
50.0%
40.0%
30.0%
20.0%
10.0%
0.0%
2.1%
3.9%
2.3%
2.3%
1.3%
5.3%
1.3%
2.8%
1.3%
3.4%
21.0%
34.0%
70.7%
48.3%
Less than 2 weeks
2 to 4 weeks
4 to 6 weeks
6 to 8 weeks
8 to 10 weeks
10 to 12 weeks
More than 12 weeks
Routine
Urgent

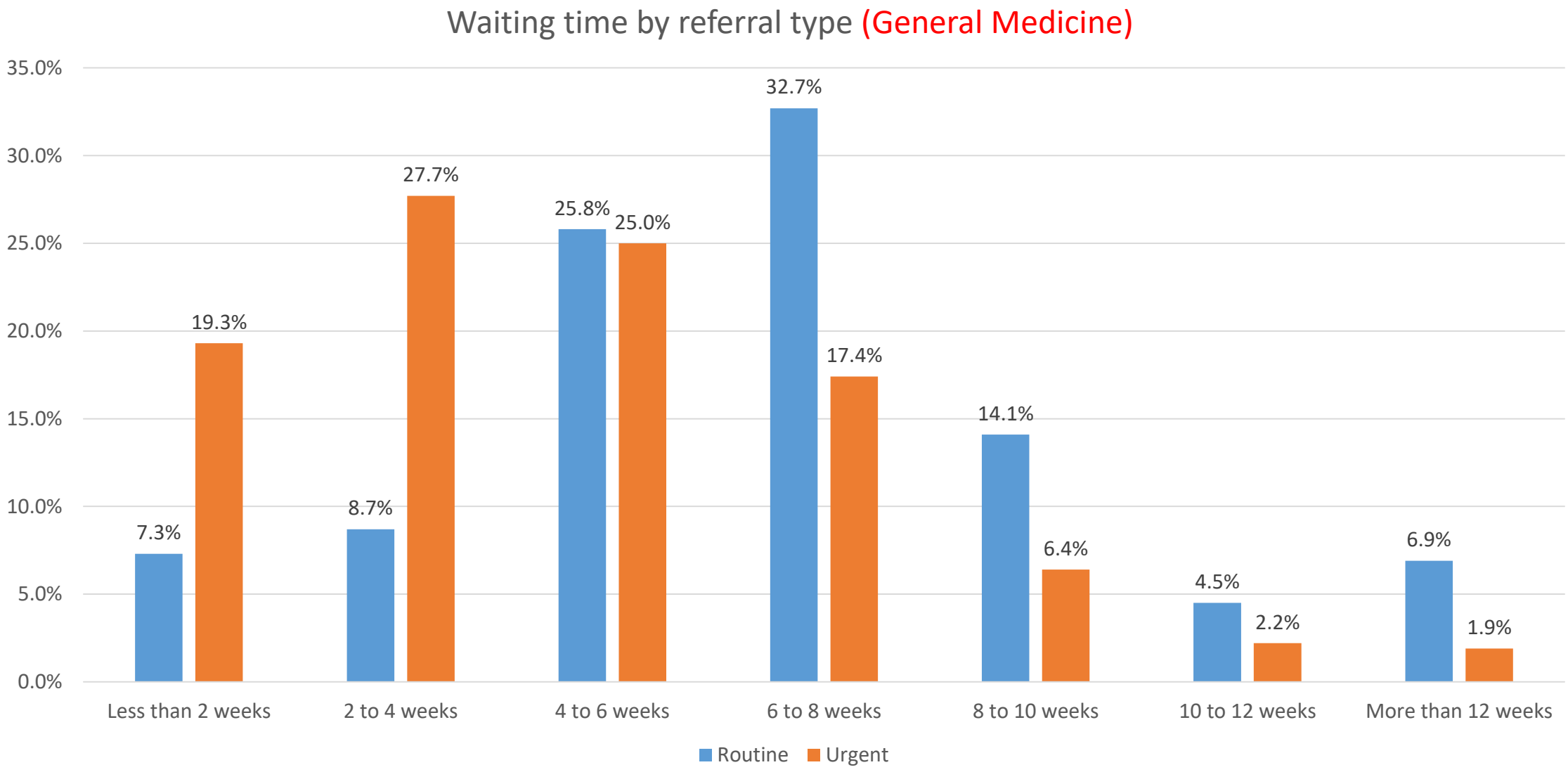
General Medicine

Routine: The median waiting time is about 45 days. Outliers are frequent, with the longest wait recorded at 335 days. Furthermore, 50% of total routine cases have a waiting time of more than 6 weeks.

Urgent: The median wait is lower, at approximately 30 days. Most urgent cases are handled within 100 days, though a few reach up to 275 days. In addition, 50% of total urgent cases have a waiting time of more than 4 weeks.



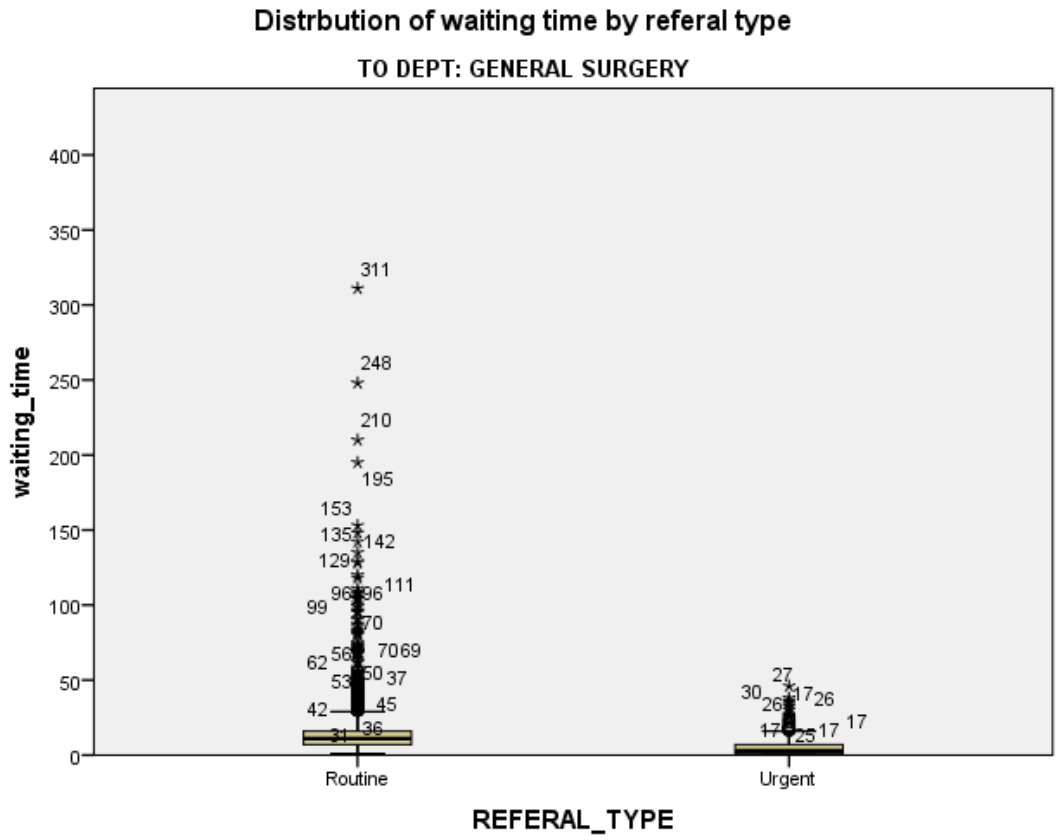


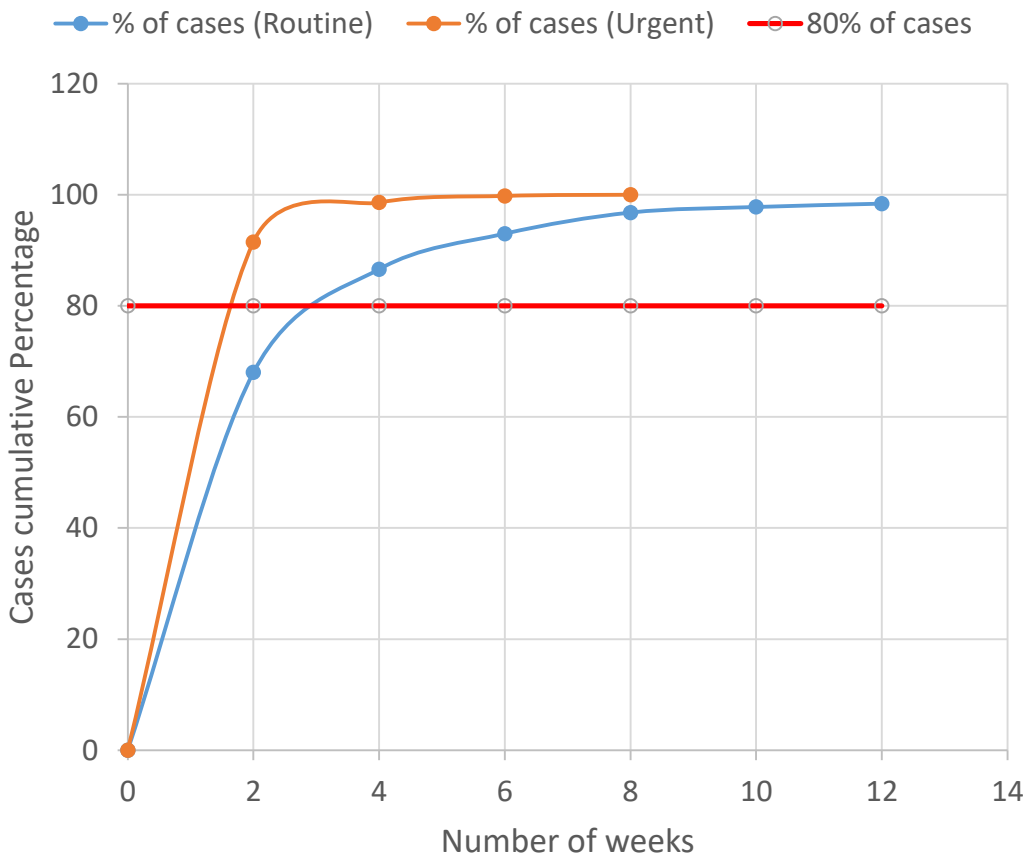

Waiting time by referral type (General Medicine)
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
7.3%
19.3%
8.7%
27.7%
25.8%
25.0%
32.7%
17.4%
14.1%
6.4%
4.5%
2.2%
6.9%
1.9%
Less than 2 weeks
2 to 4 weeks
4 to 6 weeks
6 to 8 weeks
8 to 10 weeks
10 to 12 weeks
More than 12 weeks
Routine
Urgent

General Surgery

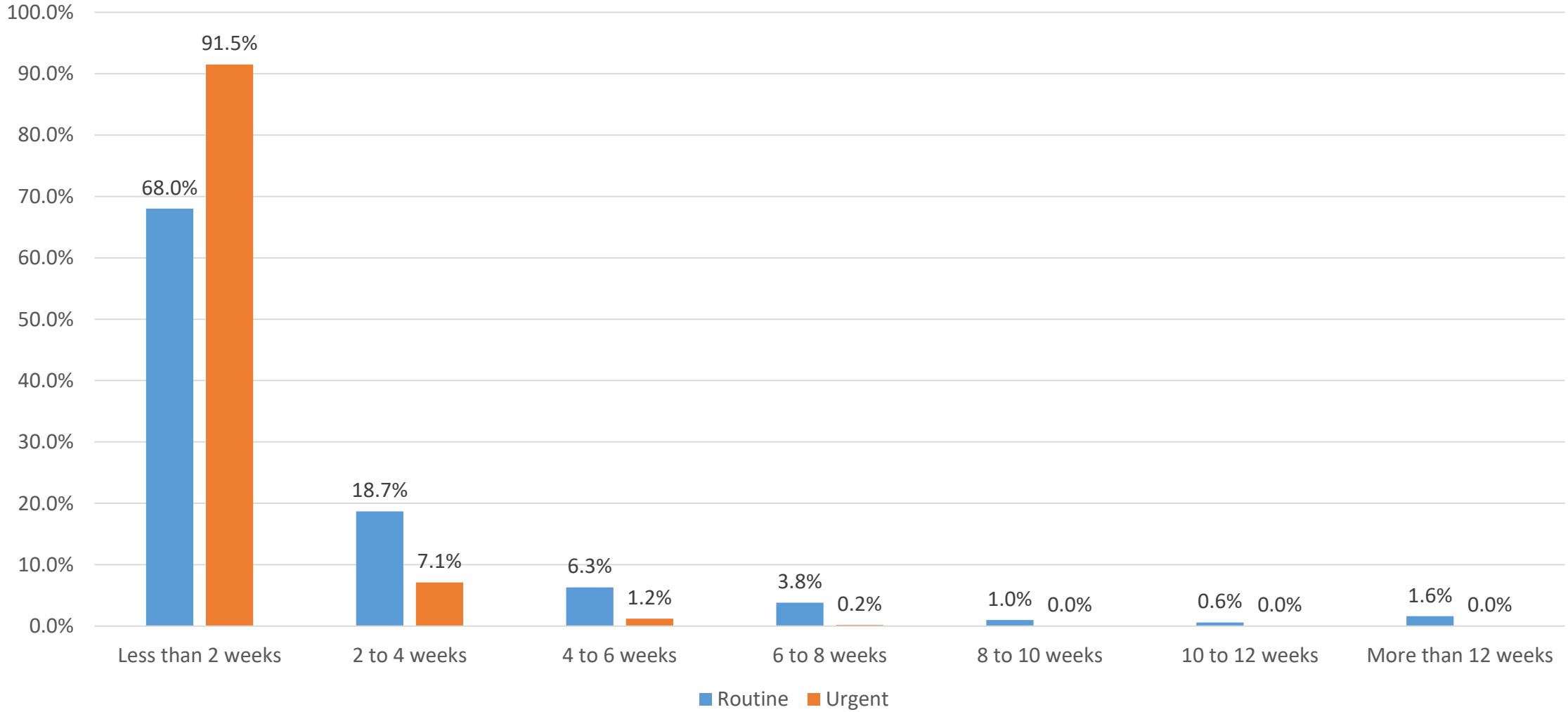
Routine: This department has a low median wait time of roughly 15 days. Despite the low average, extreme outliers exist, with one patient waiting 311 days. Further, 80% of total routine cases have a waiting time of less than 3 weeks.

Urgent: The performance here is very efficient. Most patients are seen almost immediately (median near 5 days), with very few outliers, the highest being only 47 days. Also, 80% of total urgent cases have a waiting time of less than 2 weeks.





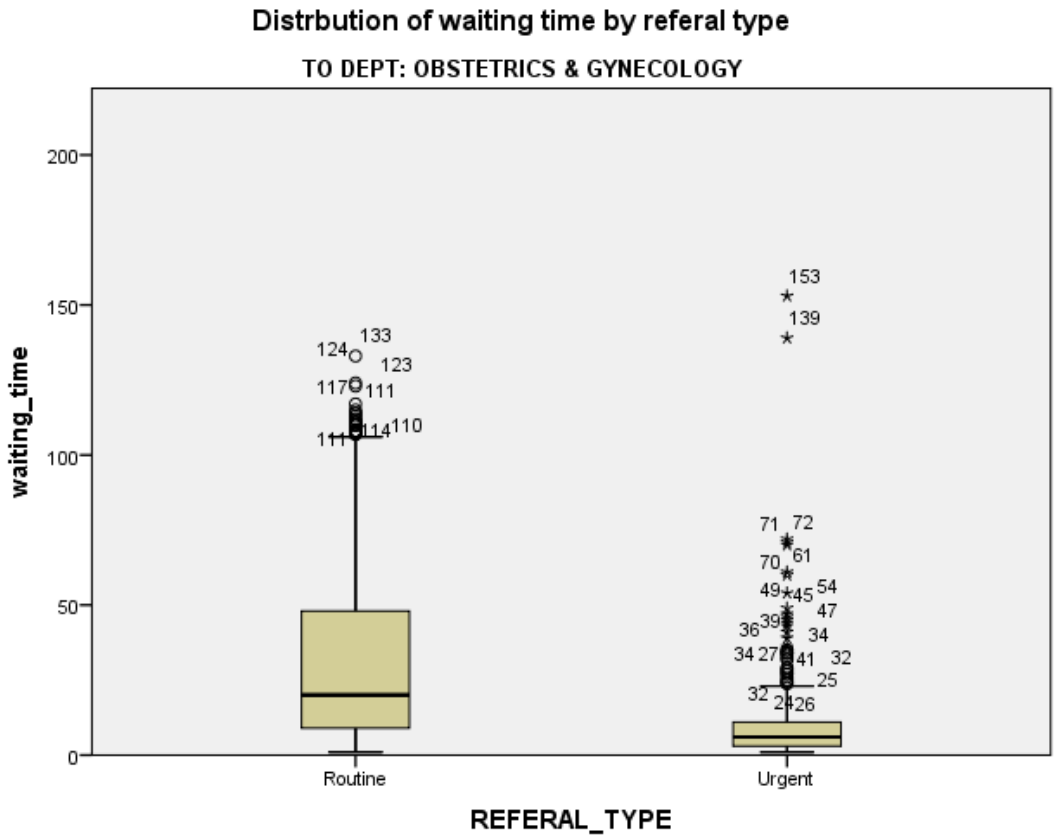
Waiting time by referral type (General Surgery)

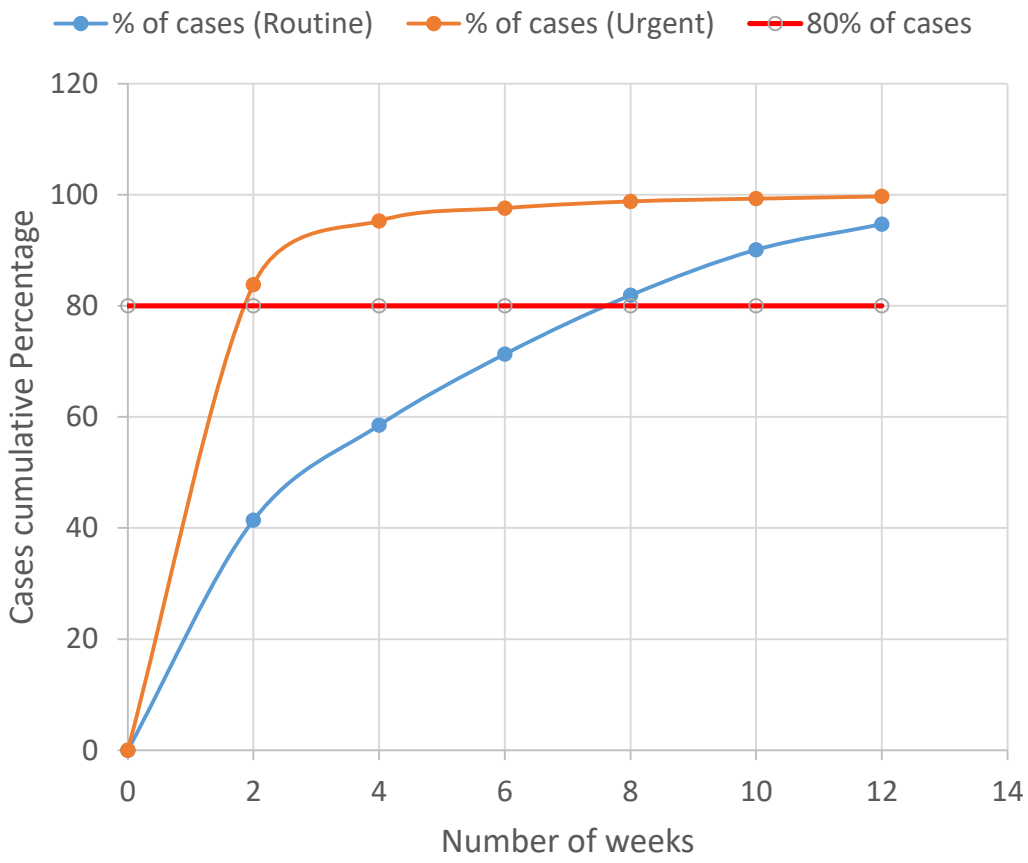


Obstetrics & Gynecology

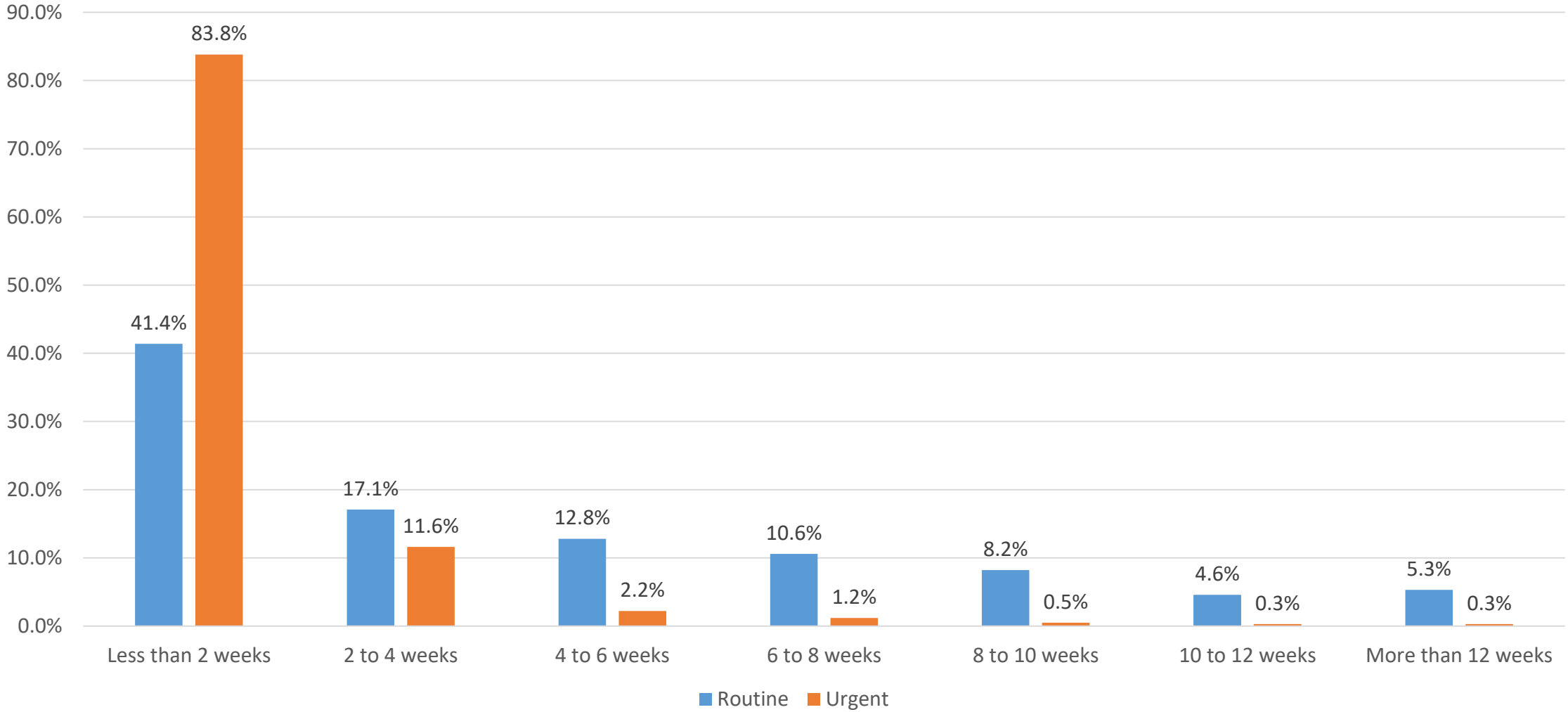
Routine: The median wait time is approximately 20 days. The distribution is skewed, with a long "tail" of outliers reaching 133 days. Furthermore, 20% of total routine cases have a waiting time of more than 7 weeks.

Urgent: Median wait time is very low (under 10 days). Interestingly, there are two distinct extreme outliers at 139 and 153 days, which are higher than the routine maximum. Besides, 80% of total urgent cases have a waiting time of approximately less than 2 weeks.





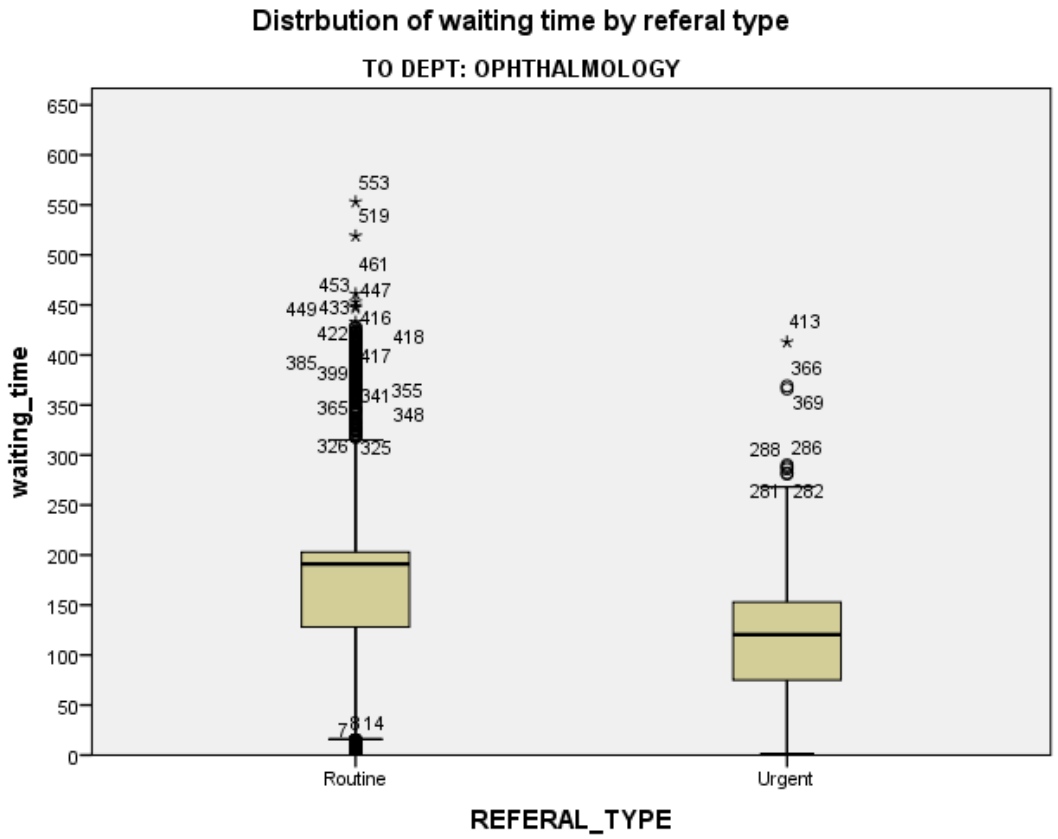
Waiting time by referral type (Obstetrics & Gynecology)

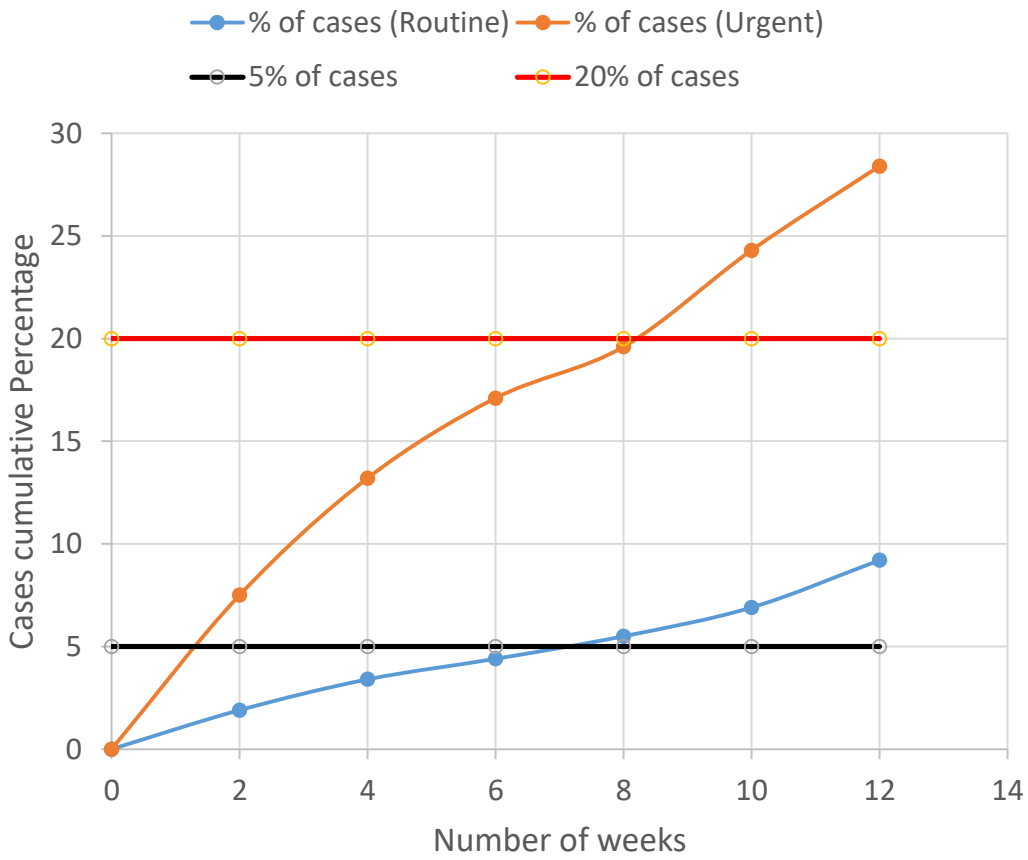


Ophthalmology

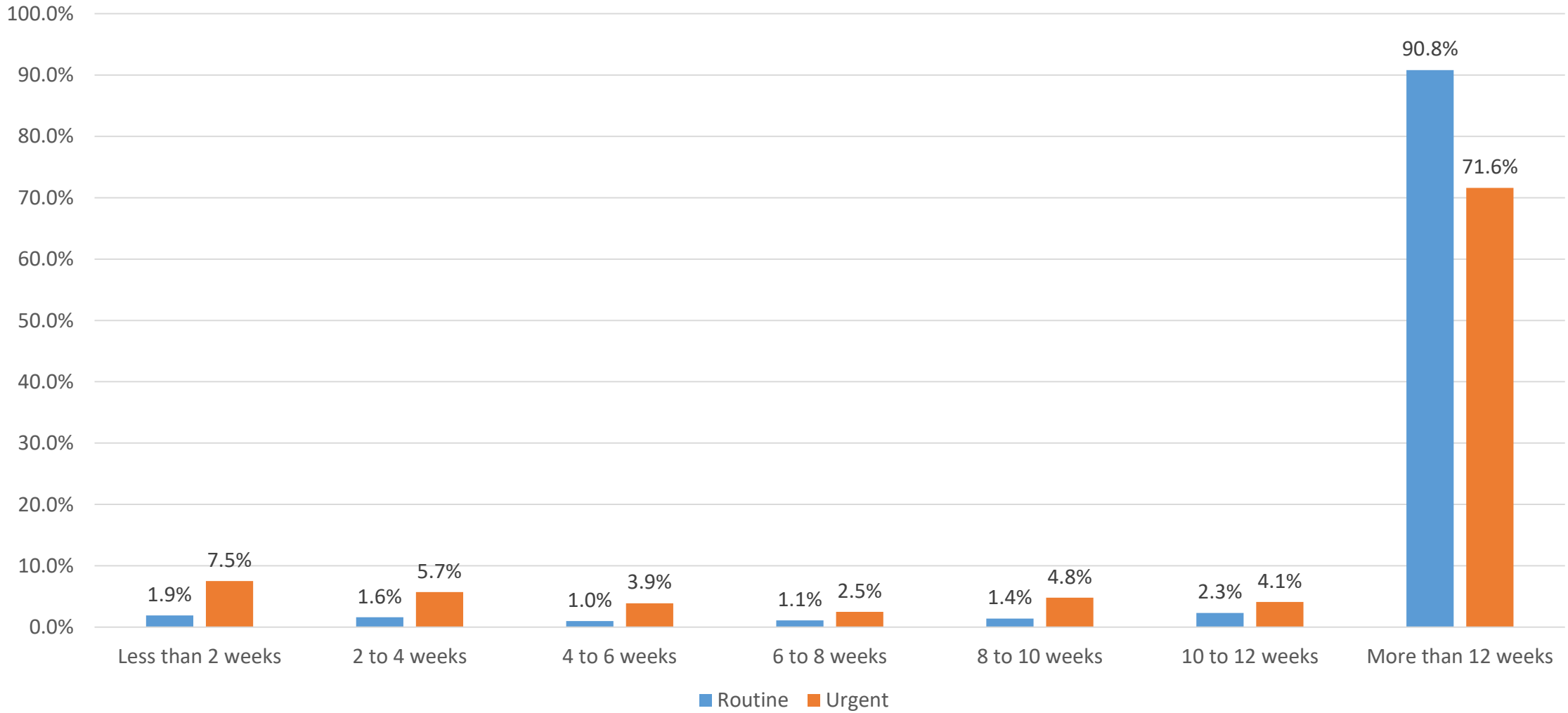
Routine: This department has some of the highest waiting times, with a median around 190 days. Outliers are extreme, peaking at 553 days (the highest in the report). Furthermore, 95% of total routine cases have a waiting time of more than 7 weeks.

Urgent: The median is lower (approx. 120 days), but still higher than the routine waits of most other departments. Urgent outliers reach up to 413 days. Moreover, 80% of total urgent cases have a waiting time of more than 8 weeks.





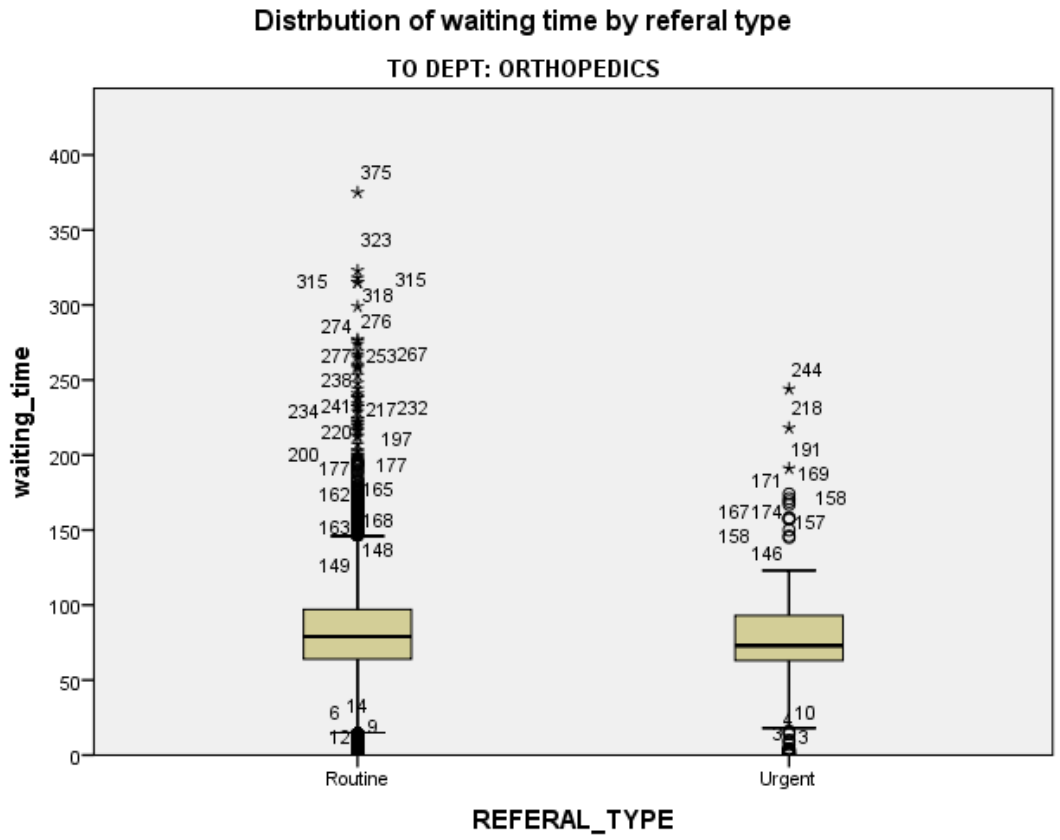
Waiting time by referral type (Opthalmology)

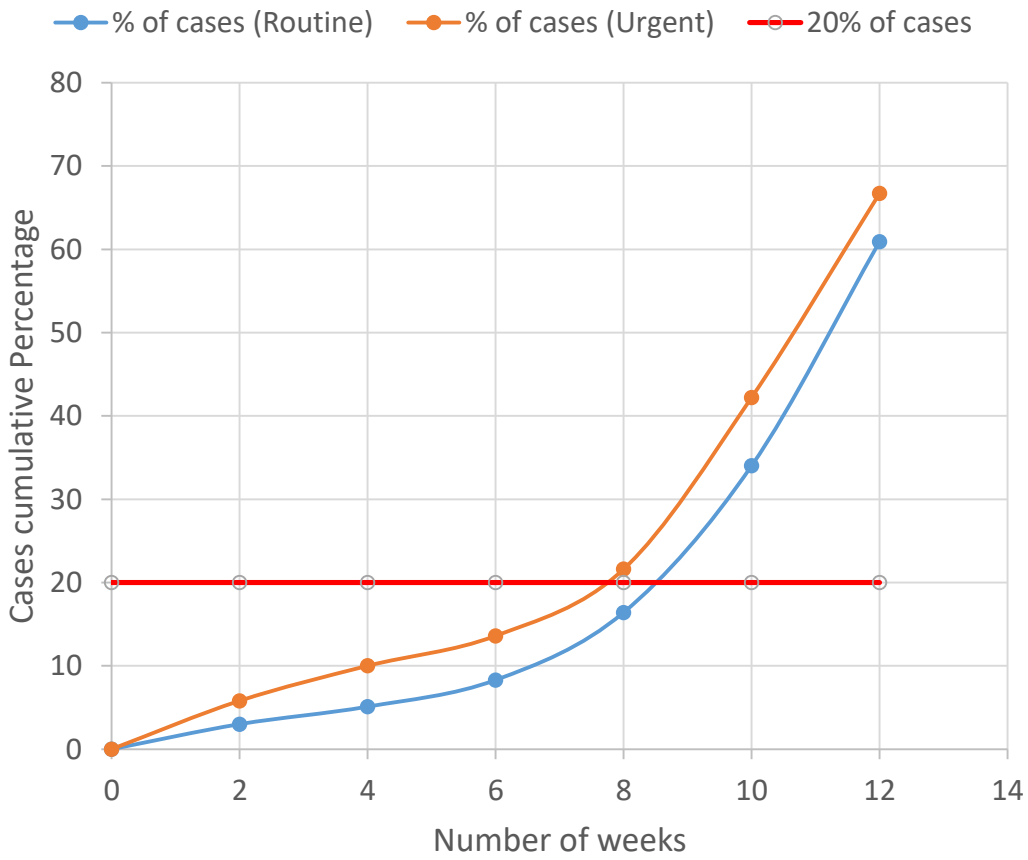


Orthopedics

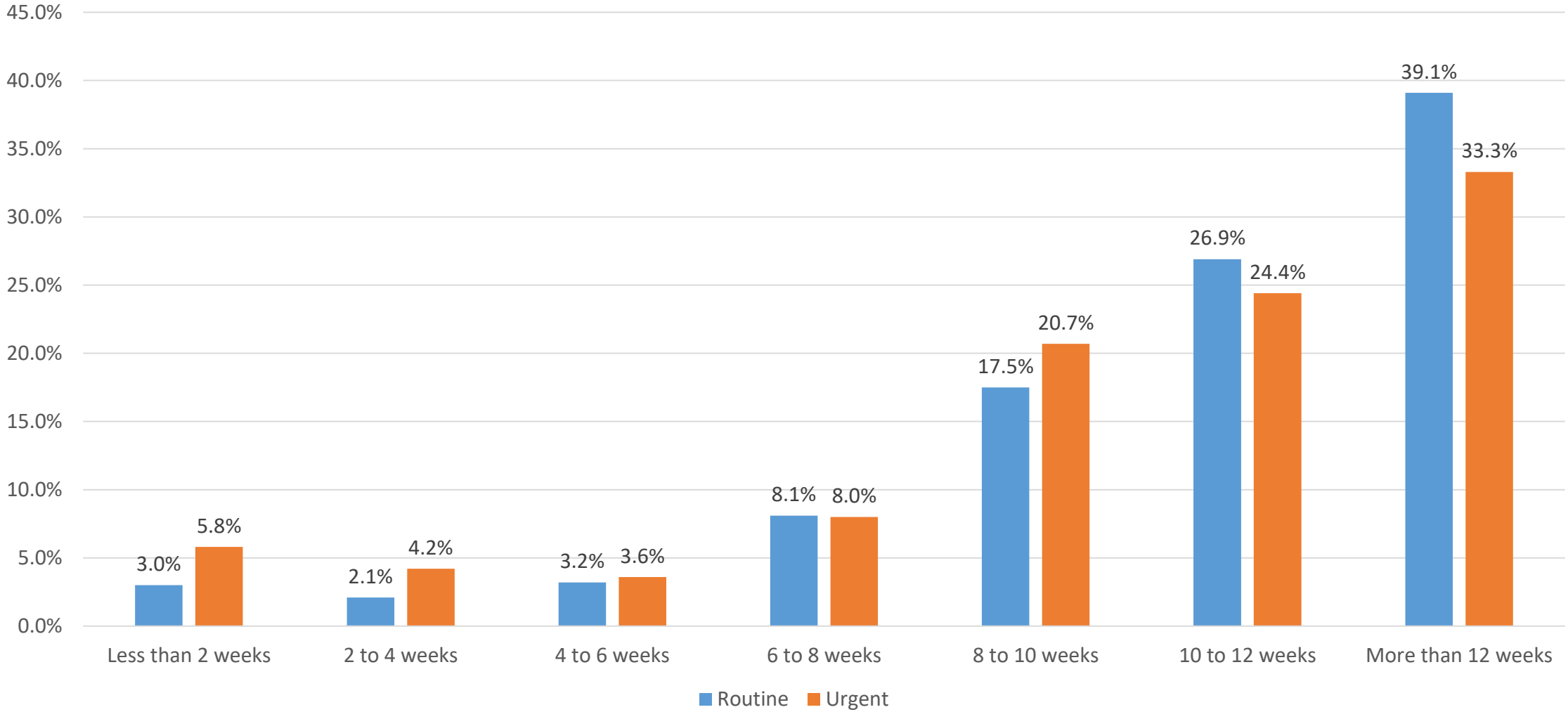
Routine: The median wait time is about 80 days. There is a very high density of outliers ranging from 150 to 375 days. Additionally, 80% of total routine cases have a waiting time of more than 8 weeks.

Urgent: The median is very similar to the routine median (approx. 75 days). Like ENT, there is significant overlap between routine and urgent wait times, with urgent outliers reaching 244 days. Furthermore, 80% of total urgent cases have a waiting time of more than 7 weeks.





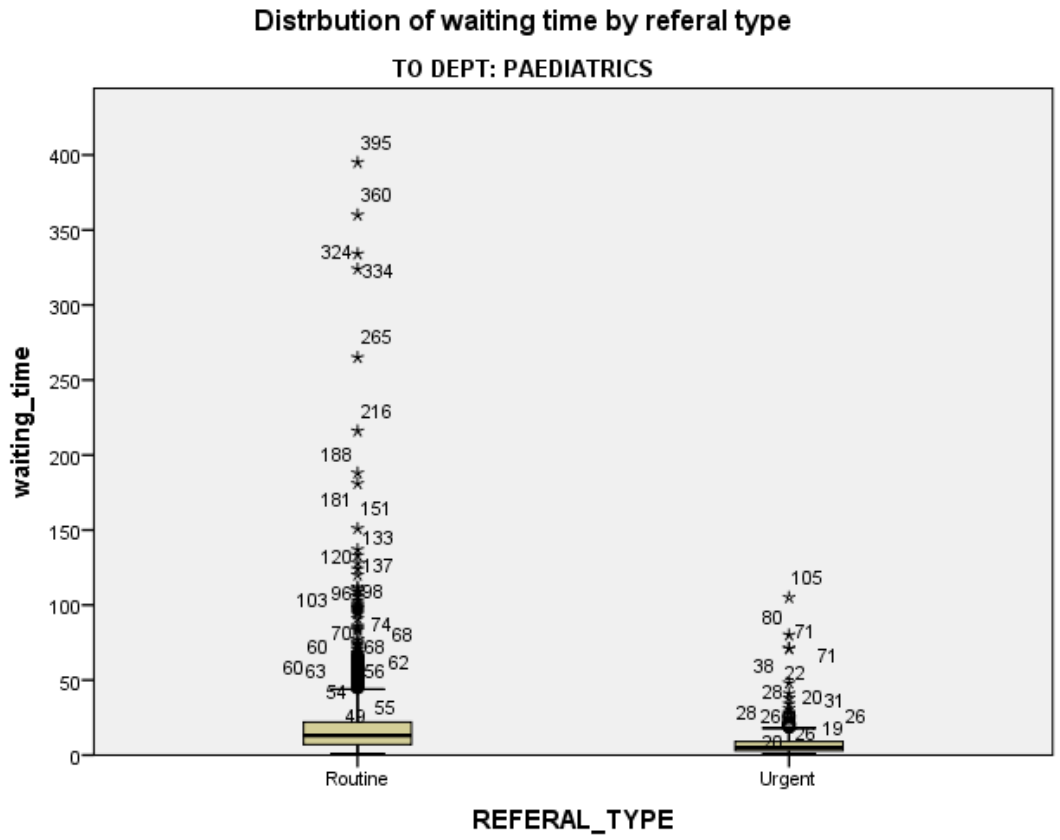
Waiting time by referral type (Orthopedics)

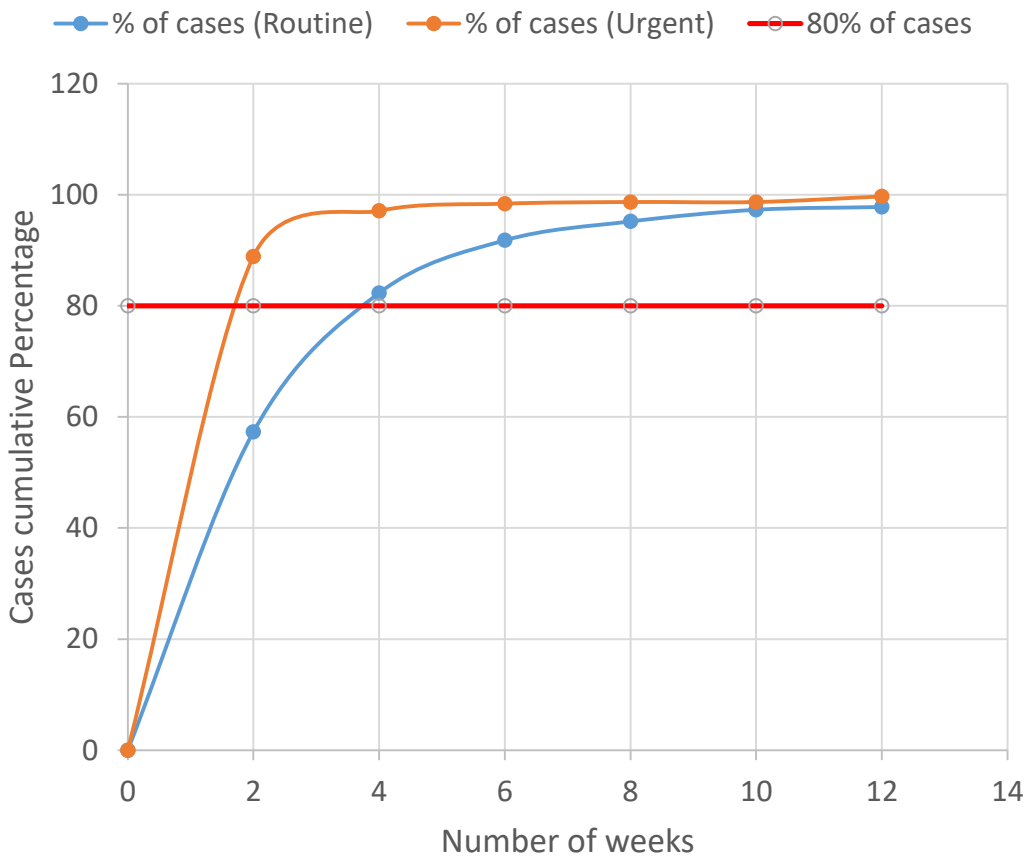


Pediatrics

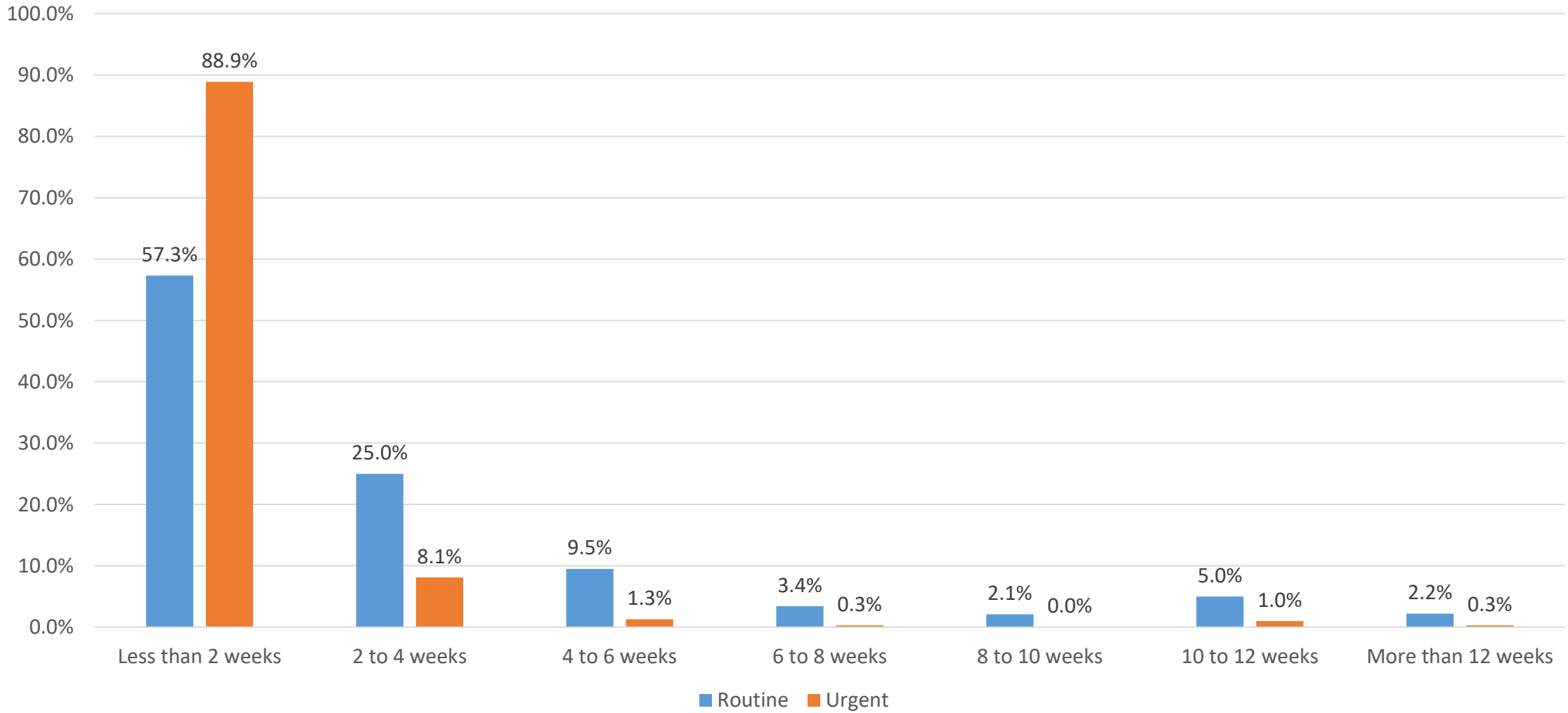
Routine: The median wait is low, around 15 days. However, the department struggles with extreme outliers, with several cases exceeding 300 days and one reaching 395 days. Also, 80% of total routine cases have a waiting time of less than 4 weeks.

Urgent: Excellent prioritization is shown here, with a median wait time of about 5 days. Most urgent cases are resolved quickly, though one outlier reached 105 days. In addition, 80% of total urgent cases have a waiting time of less than 2 weeks.

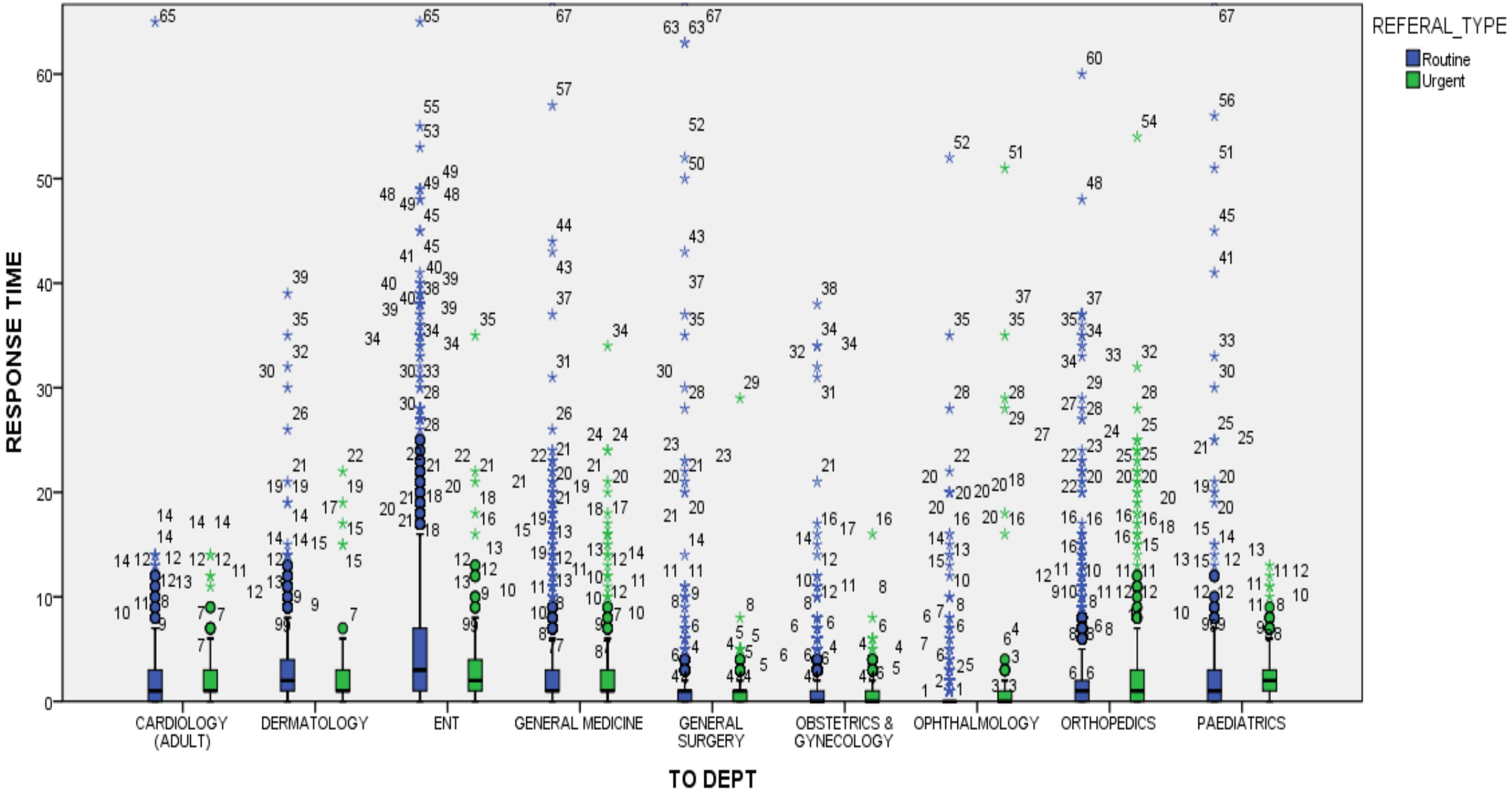




Waiting time by referral type (Paediatrics)



Section 3
Response time


REFERAL_TYPE
Routine
Urgent
RESPONSE TIME
0
10
20
30
40
50
60
CARDIOLOGY (ADULT)
DERMATOLOGY
ENT
GENERAL MEDICINE
GENERAL SURGERY
OBSTETRICS & GYNECOLOGY
OPHTHALMOLOGY
ORTHOPEDICS
PAEDIATRICS
TO DEPT

The provided box plot illustrates the Response Time (the duration between a patient's referral and the decision to approve or otherwise handle the case) for various departments at Ibra Hospital.

General Observations

Referral Types: Across all departments, data is split between Routine (blue) and Urgent (green) referrals.

Median Response: For most departments, the median response time (the horizontal line within the box) is very low, often appearing near the zero-to-ten-day mark. This suggests that 50% of decisions are made relatively quickly.

Outliers: Every department shows a significant number of outliers (individual points above the whiskers), representing unusually high response times compared to the typical case.

Summary of Efficiency

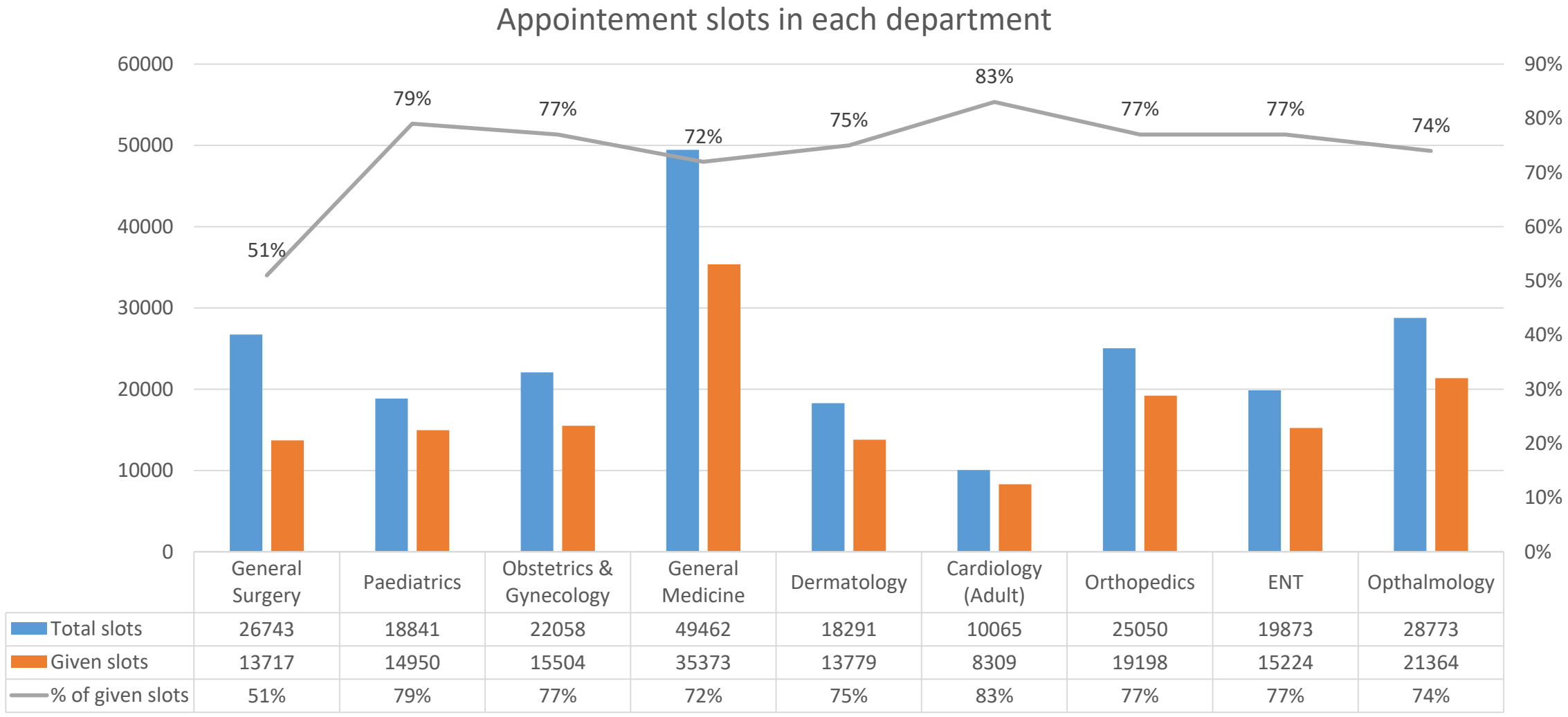
Urgent vs. Routine: Consistent with healthcare priorities, Urgent referrals (green) generally have shorter boxes and fewer extreme outliers than Routine referrals (blue), indicating they are processed more rapidly.
Consistency: Departments like Dermatology and General Medicine show tighter distributions, meaning their response times are more predictable, despite a few outliers reaching 110 days in General Medicine.

Service Gaps: The presence of extreme outliers (e.g., 283 days in ENT) highlights specific areas where care coordination may need improvement to reduce the risk of complications for those patients

Department	Interpretation of Response Time
ENT	Shows the highest extreme outliers for Routine referrals, with one case reaching **283 days** and another at **181 days**. The Urgent cases are more tightly clustered but still show delays up to 35 days.
Ophthalmology	Displays a wide spread of outliers for Routine referrals, peaking at **178 days**. Interestingly, Urgent referrals in this department also show significant outliers (up to **97 days**).
Paediatrics	Routine referrals show a notable outlier at **129 days**. Urgent referrals are handled much faster, generally staying below 30 days.
General Surgery	Most cases are resolved quickly, but Routine referrals have a cluster of outliers between **60 and 121 days**.
Cardiology (Adult)	Features one of the more efficient distributions, though a Routine outlier exists at **65 days**.
Obstetrics & Gynecology	Shows very low median times, but Routine outliers reach up to **102 days**.
Orthopaedics	Routine response times frequently deviate from the median, with several outliers between **60 and 94 days**.

Section 4
Appointment slots

This graph shows the total available appointment slots and the number of slots allocated to patients across different departments. Overall, most departments utilized between about 72% and 83% of their available appointment capacity. Adult Cardiology had the highest utilization rate at 83%, while General Surgery showed the lowest utilization at around 51%. General Medicine had the largest number of appointment slots, reflecting the high demand for services in that department. These results suggest that while most departments are using a large proportion of their capacity, some variation exists that may indicate opportunities to improve scheduling efficiency.



	General Surgery	Paediatrics	Obstetrics & Gynecology	General Medicine	Dermatology	Cardiology (Adult)	Orthopedics	ENT	Opthalmology
Total slots	26743	18841	22058	49462	18291	10065	25050	19873	28773
Given slots	13717	14950	15504	35373	13779	8309	19198	15224	21364
% of given slots	51%	79%	77%	72%	75%	83%	77%	77%	74%

Section 5
Conclusion

Report Conclusion

The report concludes that while the referral system at Ibra Hospital demonstrates effective prioritization of urgent cases and generally efficient response times, waiting time performance remains inconsistent across departments.

Key concerns include:

- Excessively long waiting times in departments such as Ophthalmology and Orthopedics.
- High variability and numerous outliers, indicating unequal patient experiences and possible system inefficiencies.
- Overlap between urgent and routine waiting times in some departments (e.g., ENT and Orthopedics), suggesting gaps in prioritization.
- These findings point to capacity constraints, high demand, and scheduling inefficiencies as likely contributing factors. Addressing these issues through better resource allocation, improved scheduling systems, and targeted interventions in high-burden departments could significantly enhance service quality and reduce waiting times.

Section
Summary tables

Waiting Time Indicators (Days)

Waiting Time (Days)	Referal Type	Indicator	Departement								
			Cardiology (Adult)	Dermatology	ENT	General Medicine	General Surgery	Obstetrics & Gynecology	Opthalmology	Orthopedics	Paediatrics
	Routine	Min	1	1	1	1	1	1	1	1	1
		Max	354	270	361	335	311	133	553	375	395
		Mean	74.7	45.6	97	49.2	16.2	30.5	172	81	19
		S.D.	34.6	25	35.3	30.1	20.2	26.6	68.6	33	27
		Median	80	39	94	46	11	20	191	79	13
		IQR	36	24	22	21	9	39	75	33	15
	Urgent	Min	1	1	1	1	1	1	1	1	1
		Max	212	168	352	275	46	153	413	244	105
		Mean	61.4	24	86.5	32.4	5.3	9.3	115	73.6	8
		S.D.	34.8	21.9	39.8	25.9	6.2	12.7	63.1	31.7	11
		Median	65	17	84	30	3	6	120.5	73	5
		IQR	52	22	19	25	6	8	78	30	6

Waiting Time Percentiles (Days)

Waiting Time (Days)	Referal Type	Percentiles	Departement								
			Cardiology (Adult)	Dermatology	ENT	General Medicine	General Surgery	Obstetrics & Gynecology	Opthalmology	Orthopedics	Paediatrics
	Routine	25%	56	32	84	36	7	9	128	64	7
		50%	80	39	94	46	11	20	191	79	13
		75%	92	56	106	57	16	48	203	97	22
	Urgent	25%	33	10	78	18	1	3	75	63	3
		50%	65	17	84	30	3	6	120	73	5
		75%	85	32	97	43	7	11	153	93	9

Referral cases (%)

Waiting Time (Days)	Referal Type	Percentage	Departement								
			Cardiology (Adult)	Dermatology	ENT	General Medicine	General Surgery	Obstetrics & Gynecology	Opthalmology	Orthopedics	Paediatrics
	Routine	less than 2 weeks	3.5	5.7	2.1	7.3	68	41.4	1.9	3	57.3
		2 to 4 weeks	6.3	11.4	2.3	8.7	18.7	17.1	1.6	2.1	25
		4 to 6 weeks	7.2	40.1	1.3	25.8	6.3	12.8	1	3.2	9.5
		6 to 8 weeks	8.7	17.9	1.3	32.7	3.8	10.6	1.1	8.1	3.4
		8 to 10 weeks	13.9	11.2	1.3	14.1	1	8.2	1.4	17.5	2.1
		10 to 12 weeks	18	9.8	21	4.5	0.6	4.6	2.3	26.9	5
		More than 12 weeks	42.4	3.9	70.7	6.9	1.6	5.3	90.8	39.1	2.2
		Total cases	1389	3693	4268	3281	1880	4001	5756	6278	1510
	Urgent	less than 2 weeks	11.1	40.3	3.9	19.3	91.5	83.8	7.5	5.8	88.9
		2 to 4 weeks	10.8	29.9	2.3	27.7	7.1	11.6	5.7	4.2	8.1
		4 to 6 weeks	10.6	15.2	5.3	25	1.2	2.2	3.9	3.6	1.3
		6 to 8 weeks	7.9	4.3	2.8	17.4	0.2	1.2	2.5	8	0.3
		8 to 10 weeks	14	6.2	3.4	6.4	0	0.5	4.8	20.7	0
		10 to 12 weeks	17.7	3.3	34	2.2	0	0.3	4.1	24.4	1
		More than 12 weeks	27.8	0.9	48.3	1.9	0	0.3	71.6	33.3	0.3
		Total cases	378	211	435	591	493	579	690	450	307


ENDS
REPORT